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Secondary instabilities in flat plat boundary layer flow for supercritical fluids

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Abstract

In an incompressible isothermal flat-plate boundary layer flow, the unstable eigenmodes known as TS waves, the related secondary modes, and their contributions to the laminar-turbulent transition have long since been investigated. Recently, Ren et al. (2019b) discovered a new unstable mode in a boundary layer flow of supercritical CO_2. The main objectives of this study are to examine the possible secondary instability related to this newly discovered mode through direct numerical simulation (DNS), analyze the nonlinear effects and resonance interactions which may come along, and finally discuss their potential effects on transition.

The DNSs are realized through an in-house code in `FORTRAN`, in which Van der Waals equation of state is used to model the real gas effects brought by the transport and thermodynamic properties of supercritical fluids. After the simulation is initialized by a laminar boundary layer flow, a blowing/suction strip at specific frequencies and amplitudes is applied to trigger instability. DNS data are collected when flow motion reaches a periodic state in the whole domain. The post-processing is then effectuated with `MATLAB` scripts which are validated by recovering the analytical results of unstable modes in incompressible isothermal boundary layer. To detect the secondary instability growth, 2D simulations are first conducted with different forcing amplitudes, and after the secondary growth is observed, the 3D configuration is used to discuss the effects of spanwise wave number.

Our results confirm the existence of secondary instability growth in the boundary layer of supercritical fluids. However, we also observe non-linear effects and interactions between higher harmonics, which are negligible in the case of incompressible and isothermal flows. Considering the large growth rate of the new mode and the significant non-linear effects, we speculate that, in the studied configuration, transition to turbulence in the boundary layer of supercritical fluids is less likely to follow the path via secondary instability.

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Nomenclature

Acronyms

CFL	Courant–Friedrichs–Lewy number
DNS	Direct Numerical Simulation
EoS	Equation of State
FFT	Fast Fourier Transformation
FT	Fourier Transformation
LST	Linear Stability Theory
SCF	Supercritical Fluid
TS	Tollmien-Schlichting
VdW	Van der Waals

Dimensionless Numbers

Ec	Eckert Number
Ma	Mach Number
Pr	Prandtl Number
Re	Reynolds Number

List of Symbols

α	Streamwise wave number
α_i	Local growth rate
β	Spanwise wave number
$\delta(x)$	Local boundary layer thickness scale
δ_{99}	The 99% boundary layer thickness
κ	Thermal conductivity
μ	Dynamic viscosity
ν	Kinetic viscosity
ω	Angular frequency of the perturbation
ρ	Density
ϑ	Specific volume
c_p, c_v	Specific heat capacity
L_x	Streamwise domain length
L_y	Wall normal domain length
L_z	Spanwise domain length
L_{sp}	Sponge length
n_x	Number of grid points in the streamwise direction
n_y	Number of grid points in the wall normal direction
n_z	Number of grid points in the spanwise direction
p	Pressure
R	Specific gas constant
S_f	Grid stretching factor
T	Temperature
u	Streamwise velocity
v	Wall normal velocity
w	Spanwise velocity
x	Streamwise coordinate
y	Wall normal coordinate
Z	compressibility factor
z	Spanwise coordinate

Superscripts and Subscripts

$(\cdot)'$	Fluctuation quantity
$(\cdot)^*$	Dimensional quantity

$(\cdot)_\infty$ Free-stream quantity

$(\cdot)_c$ Critical value

$(\cdot)_i$ Imaginary part

$(\cdot)_r$ Real part

$(\cdot)_w$ Wall value

$(\cdot)_{pc}$ Pseudo-critical value

$\bar{(\cdot)}$ Time-averaged quantity

$\hat{(\cdot)}$ Fourier-transformed quantity

Chapter 1

Introduction

This chapter introduces important concepts and previous research progress, including laminar-turbulent transition, stability theory in flat-plate boundary layer flow, special features of supercritical fluids (SCF) and their influence on hydrodynamic behaviors. We end this chapter by a presentation on the objectives and work plane of this study.

1.1 Transition to turbulence

Laminar-turbulent transition is triggered by disturbances in the flow. This is a complex process which can follow different routes even with the same type of flow. Morkovin (1969) described several possible routes for transition, shown in figure1.1. In his classification, transition route depends on the disturbance level.

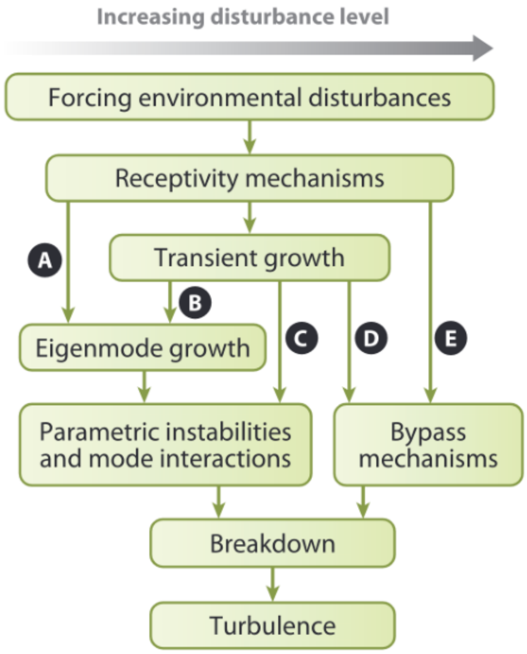


FIGURE 1.1: Possible routes of laminar to turbulent transition (Morkovin, 1969).

After the appearance of disturbances in the flow, marked as receptivity mechanisms stage in the figure, if the disturbances is small, the transition follows a modal route, which involves the growth of unstable modes. Path A, B, C are all related to modal instability growth in the flow, including eigenmode growth and/or growth due to resonance interactions. For path A, the disturbance level is small and only unstable eigenmodes start to grow. When their amplitudes reach to a certain level, mode interactions happen, including the secondary instability growth which is the mean subject of this study. When the initial perturbation becomes greater, the transient growth of stable mode can also complement eigenmode growth or directly trigger mode interactions, marked by routes B and C, respectively. When the external disturbances are strong, transition can also happen via *bypass transition*, which is the special mechanism in routes D and E. This notion is first introduced by Morkovin (1969) himself. The word *bypass* signifies that this is a path via non-modal growth mechanism.

The classification of Morkovin (1969) is general. Within each potential route, there are also several ramifications, usually identified by how flow patterns change during transition. Apart from the initial disturbance level, other factors can also decide through which path and scenario the transition happens. Schmid and Henningson (2001b) listed some of the factors: the initial disturbances spectrum, the development speed and the required energy to reach a turbulent state for each scenario.

In this study we focus on the modal transition path A. To be more specific, the modal transition in a boundary layer flow of SCF. While SCF introduces real gas effects in thermodynamic and transport properties of the flow, we would like to introduce firstly the notions and research background of stability and transition in idealized flat-plate boundary layer flow.

1.2 Stability and modal transition to turbulence in a flat-plate boundary layer flow

For flat-plate boundary layer flow, the modal transition via instabilities has its own features. Figure 1.2 shows a sketch from experiments, which illustrates transition stages in this specific case. Here, the eigen-

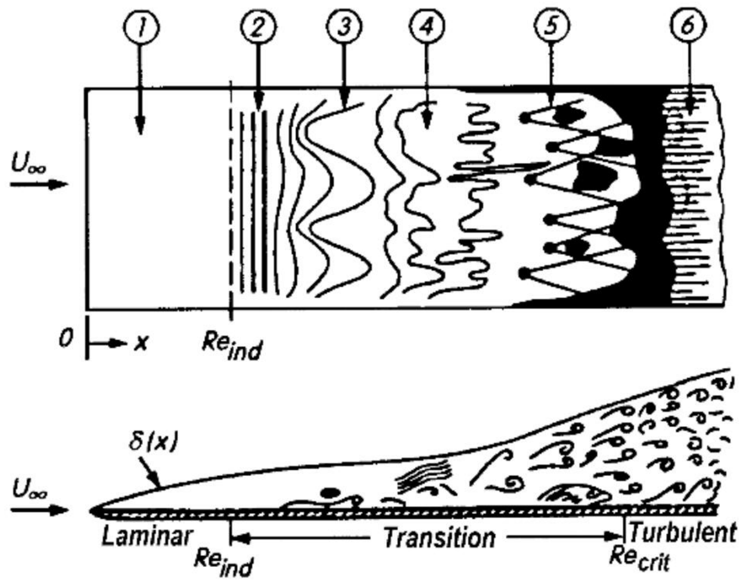


FIGURE 1.2: Sketch of laminar-turbulent transition in boundary layer on a flat plat. (1) stable laminar flow, (2) unstable Tollmien-Schliching waves, (3) 3D waves and vortex originated from secondary instability (Λ structure) (4) vortex decay (5) formation of turbulent spots (6) fully turbulent flow. Figure from Schlichting and Gersten (2017).

mode growth stage in figure 1.1 is related to Tollmien-Schlichting waves (TS waves). As the boundary layer flow develops downstream, secondary instabilities start off, then intermittent structures appear as 3D waves and vortex, and ultimately leads to fully developed turbulence. In this section, the notions of TS waves and secondary instabilities in a flat-plat boundary layer are introduced. As the ultimate breakdown to turbulence itself is not the main focus of this study, it is not described in detail.

1.2.1 Linear unstable modes: TS waves

Linear stability is related to linear stability analysis, which assumes that the magnitude of perturbations is infinitesimal compared to the base flow, so the system of equation can be linearized to study the behaviors of these perturbations. More precisely, the perturbations appear in form of waves, and their amplitudes whether change in time (temporal evolution) or in space (spacial evolution).

Following the idea of Prandtl (1921) about viscous instability mechanism, Tollmien (1930) and Schlichting (1933) first predicted the oscillations in a Blasius boundary layer, which are named after them as *Tollmien-Schlichting waves*. Later, the first experimental demonstration was carried out by Schubauer and Skramstad (1947), proving their prediction. Since then, countless studies followed their lead, continuing the investigations into the instabilities in flat-plate boundary layer flow.

One of the facts that make stability analysis difficult for boundary layer flow is the growth of boundary layer thickness as the flow develops downstream. This means that the flow is not strictly parallel. *Local stability analysis* (or *parallel stability analysis*) can be used to simplify this specific problem. There are mainly two assumptions: the characteristic length of the growth of boundary layer thickness is much greater than that of the perturbations, and the perturbations rapidly adapt to the new local condition when advected downstream. With these assumptions, the wall normal velocity can be neglected. In local stability analysis, Orr–Sommerfeld equation, an eigenvalue equation describing modal behavior in 2D shear flow, can be used with the local streamwise velocity profile $U(y)$ at a fixed streamwise position x, where y is the wall-normal direction. The eigenmodes found in this way are parametric functions of x (Charru, 2011).

By contrast, *non-parallel analysis* takes normal velocity effects into account, and the growth is dependent on the criterion of instability. Gaster (1974) studied non-parallel effects and his results were confirmed by direct numerical simulations of Fasel and Konzelmann (1990). Figure 1.3 shows the theoretical neutral amplification curves derived by different methods. Instabilities are amplified inside the horn shape, and are damped otherwise. The different neutral amplification curves are found by parallel analysis or non-parallel analysis. Non-parallel analysis with kinetic energy criterion gives a larger unstable range, but the integral of $\bar{u}^2$ criterion gives similar curve as parallel analysis. When the dimensionless frequency is relatively small, the differences between different curves are negligible. Furthermore, as Charru (2011) pointed out, non-parallel effects are of little importance for 2D waves and negligible pressure gradient.

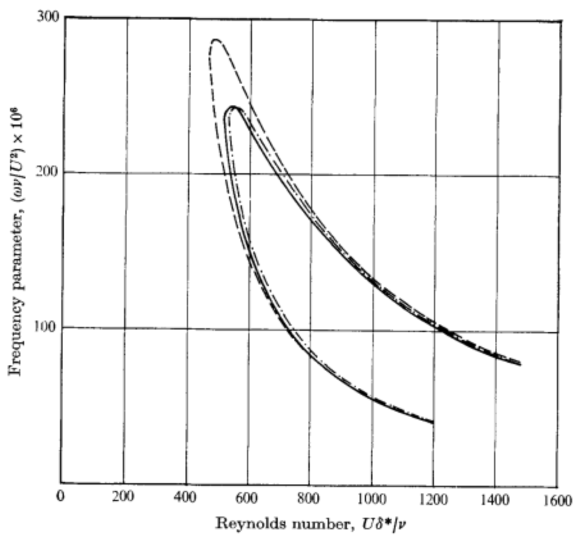


FIGURE 1.3: Neutral Amplification curves derived by different methods: parallel analysis (—); non-parallel analysis with kinetic energy criterion (- - -) and integral of $\bar{u}^2$ (-·-·). Figure from Gaster (1974).

1.2.2 Beyond unstable eigenmode growth

When the amplitude of TS waves grows to certain degree, the linear stability assumption no longer holds and non-linear effects enters into the problem. To elucidate properties of these fluid flows, nonlinear theory of hydrodynamic stability was proposed and has been developing since then. There a large amount of literature on this subject, presenting various methods to capture the non-linear development of flow disturbances, for example the work of Trefethen et al. (1993), Grossmann (1996). One of methods that can deal with non-linear effects is the energy method, which reduces non-linear terms using the fact that kinetic perturbation energy cannot be increased or decreased by non-linear terms. There is also multi-scale analysis, with which we are able to derive weakly non-linear equations assuming the amplitude of higher order disturbances varies much slower than the wavy behavior of the first order disturbances. It is also referred as *Landau's equation* (Landau and Lifshits, 1954; Schmid and Henningson, 2001a) or *Landau's approach* (Yaglom and Frisch, 2012). However, this approach also has its limitations, it traces mainly one mode, and cannot well characterize interactions between different modes. To deal with this, researchers developed more sophisticated methods, including the generally called *multi-modes analysis*. One of methods in multi-modes analysis is based on the resonant interaction of three waves of finite and comparable amplitude. Craik made great contributions to this topic and one of his studies showed that, under given conditions, there exist a resonant triad, consisting pairs of oblique waves of small amplitudes and a 2D wave of larger but finite amplitude which keeps the same as it enters the primary equilibrium solution (Craik, 1975; Yaglom and Frisch, 2012). This is closely related to the secondary instability mechanism that is presented in the coming part.

1.2.3 Secondary instabilities

Secondary instability theory is a kind of stability analysis applied to a new periodic base flow composed of the original based flow and the quasi-steady primary mode with finite amplitude. Depending on the type of secondary instabilities and the type of modified base flow on which they grow, there are several different types of secondary instabilities. For flat-plate boundary layer flow, there is for example the streak secondary instabilities which grow on the streaks linearly produced by streamwise vortices. As TS waves are the typical unstable modes, the most studied types of secondary instabilities are based on TS waves (Schmid and Henningson, 2001b). Herbert studied thoroughly about secondary instabilities in shear flows (Herbert, 1983, 1985, 1988; Herbert et al., 1987). His theoretical analysis matches well to experimental and numerical results before and after his study (Klebanoff et al., 1962; Krist and Zang, 1987; Rist and Fasel, 1995; Sayadi et al., 2013). More precisely, Herbert (1988) used Floquet analysis and classified secondary instabilities of TS waves into 3 types, fundamental modes, subharmonic modes and detuned modes, based on the streamwise wave number α. The fundamental modes are associated to the primary resonance in the Floquet system and are first observed by the experiences of Klebanoff et al. (1962), hence it is also called *K-type* secondary instability. The subharmonic modes originate from principal parametric resonance and it is named after Herbert as *H-type*. The detuned modes are related to combination resonance. As we consider the spatial growth of unstable modes, the frequency of fundamental modes is the same as the primary instabilities while the frequency of subharmonic modes is half of that.

Because of their different wave numbers, when the amplitudes of H-type and K-type grow large enough to raise 3D structures, different patterns can be observed. As mentioned before, Λ structures appear due to the growth of unstable modes, but they are arranged differently, depending on the type of secondary instability. Figure 1.4 shows a sketch of the arrangement of Λ structures. For the fundamental modes (K-type), the structures are aligned (left side of the figure), while the structures related to subharmonic modes (H-type) are staggered (right side of the figure). These patterns can also be modelled by the secondary instability analysis and we explain this more precisely in the next chapter.

While research on stability and modal transition in boundary layer flow began with incompressible isothermal fluids, later studies took more elements into consideration, enriching the physics of the flow behavior and transition. For example, Lees and Lin (1946), Mack (1984) investigated thermal effects and

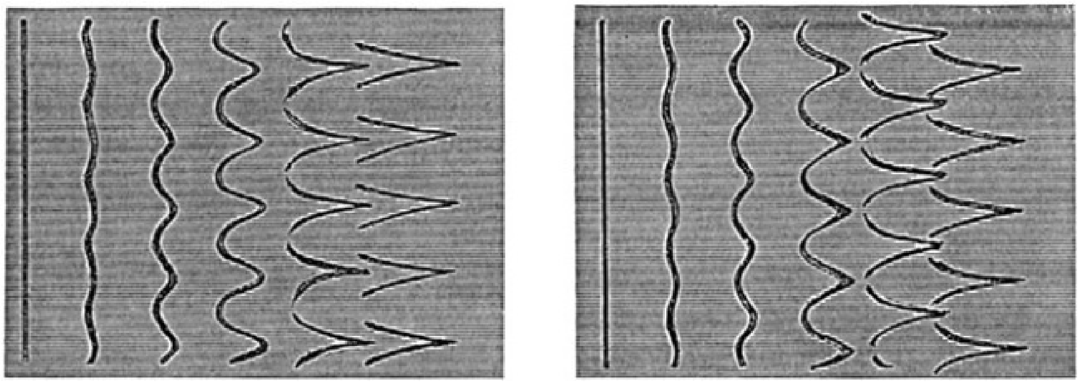


FIGURE 1.4: Streaklines in the Blasius boundary layer. The flow is from left to right. Left: Ordered (peak-valley) arrangement of Λ vortices caused by a fundamental mode. Right: Staggered arrangement of Λ vortices, caused by a subharmonic mode. Figure from Saric and Thomas (1984).

compressibility effects on the instabilities, Stuckert and Reed (1994) considered high-temperature chemical reactions in stability analysis. The main subject in this study, fluids near critical point, can also bring special features to stability and transition behaviors of flat-plate boundary layer flow. In the next section, we introduce supercritical fluids (SCFs) with their special properties, including an overview of studies about hydrodynamics and stability of SCF.

1.3 Supercritical fluids

Substances exist in different phases: solid, liquid and gas. Usually, at a given pressure, phase-change between saturated vapor and saturated liquid happens with constant temperature while density varies, known as the boiling process. When the pressure becomes higher, the saturated liquid and saturated vapor states become identical at critical point. The temperature and pressure at this point are defined respectively as *critical temperature* and *critical pressure*. A substance is called *supercritical fluid* (SCF) when its temperature or pressure becomes greater than its critical value. Above the critical point, the boiling process no longer exists and phase-change becomes subtle. For a SCF at a given pressure, the pseudo-critical point is where the specific heat C_p reaches its maximum, and the *Widom line* is a line formed by pseudo-critical points at different pressures (Sciortino et al., 1997; Raju et al., 2017). The SCF behaves as a liquid-like high-density fluid below pseudo-critical temperature, and a vapor-like low-density fluid while above. The fluid properties, including transport properties and thermodynamic properties, change significantly within a narrow temperature range while crossing the pseudo-critical point at a supercritical pressure. We call it *transcritical condition* when the temperature is in this range. A boundary layer flow in which transcritical condition exits is named as a transcritical boundary layer (Kawai, 2016).

1.3.1 Thermodynamic properties

Carbon dioxide (CO_2) is selected as the substance to be studied. Figure1.5 shows the $P-\vartheta$ diagram for CO_2. The constant pressure (80 bar) line cross the Widom line passing through the pseudo-critical point. The contour of compressibility factor Z which is defined by

$$Z = \frac{p}{\rho R T} \tag{1.1}$$

is also presented in the figure. For ideal gas $Z = 1$, and smaller Z indicates stronger the non-ideal-gas effects. With this factor, figure 1.5 indicates the large non-ideal effects in the vicinity of Widom line.

The thermodynamic and transport properties of CO_2 at $p^* = 80$ are shown in figure1.6, the values of

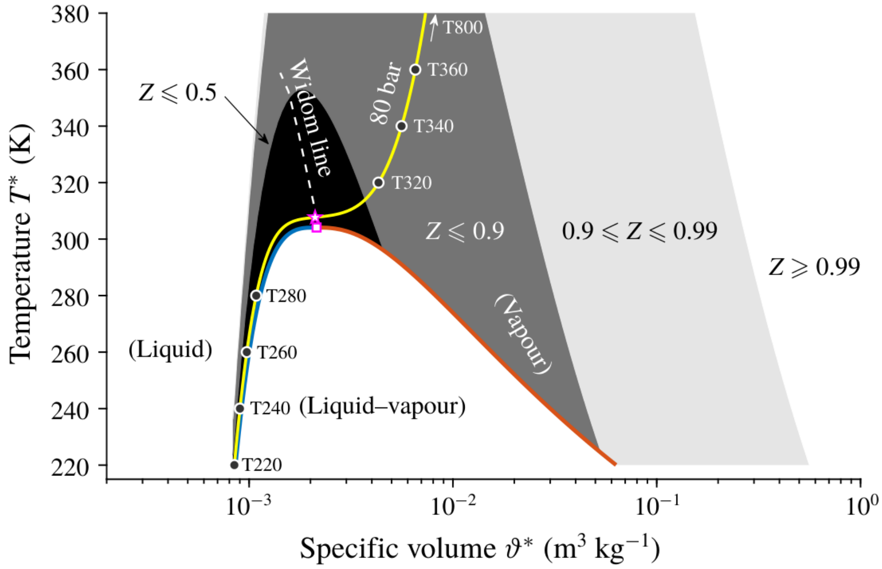


FIGURE 1.5: T-ϑ diagram for CO_2, showing critical point (magenta square), pseudo-critical point (magenta pentagram), Widom line (white dashed line), two saturation curves (blue and red thick lines) and an isobar of 80 (yellow line). The shaded area shows the contour of compressibility factor Z. Figure from Ren et al. (2019b).

REFPROP come from measurements. Van der Waals, Peng-Robinson and Redlich–Kwong are equation of state models especially established to recover $p - V - \rho$ relation for real gas. It is clear that the significant property change in pseudo-critical transition region cannot be captured by ideal gas model, thus better model must be chosen when non-ideal gas effects enters into the problem.

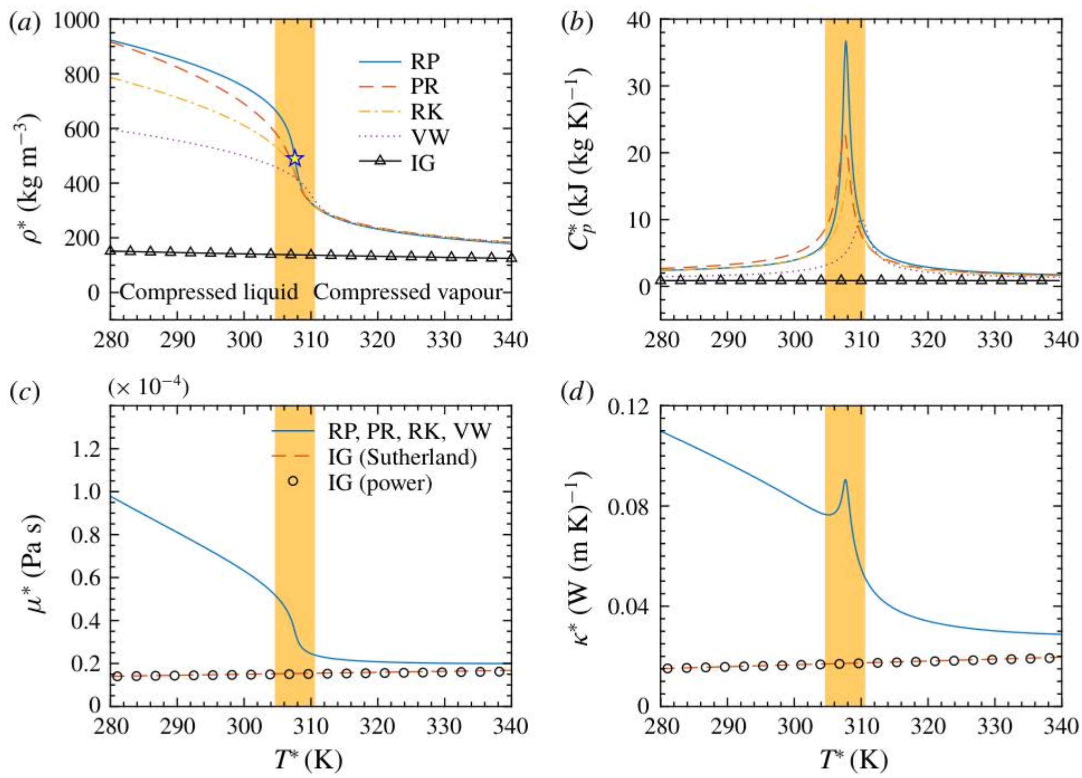


FIGURE 1.6: Thermodynamic and transport properties of CO_2 at $p^* = 80$ bar with different models: ideal gas (IG), VW(Van der Waals), Peng-Robinson(PR), Redlich–Kwong(RK), REFPROP tables (RP); REFPROP tables are used for transport properties for the real gas EoS models. The colored area indicates the pseudo-critical transition. Figure from Ren et al. (2019a).

1.3.2 Hydrodynamics of supercritical fluids

With its special features, using SCFs in many industrial applications can optimize the process and improve efficiency. One of the examples is the supercritical generator in power plants (Cheng and Schulenberg, 2001; Liu et al., 2019). Because gas turbines using SCF as working fluid have very high reachable efficiency at moderate turbine inlet temperature (Pecnik et al., 2012), supercritical generator is an attractive new option for medium capacity energy conversion. Another application is related to refrigerating or cooling system. In the study of Pizzarelli et al. (2015), it is more precisely the cooling system in rocket engine. While technologies using SCFs are under active development, researches about the hydrodynamics of SCFs are also progressing rapidly. As most of these applications favor better heat transfer, the involved flow is often at high Reynolds number conditions thus turbulent. For this reason, a lot of studies focus on turbulent behaviors and turbulence modelling of SCFs. Yoo (2013) reviewed heat transfer in SCFs flows. Pecnik and Patel (2017) proposed an alternative formulation of the turbulent kinetic energy equation in channel flows. Several studies used direct numerical simulation (DNS) to investigate the statistics of turbulence in jet (Sharan and Bellan, 2021), pipe (He et al., 2021) and also flat plat boundary layer flows (Kawai, 2016).

Compared to turbulence related topics of SCFs, less studies dedicate to understand stability and modal transition to turbulence in SCF flows. It is not until 2019 that a new unstable mode was discovered in flat-plate boundary layer flow of SCFs, to which our investigations are closely related. In the following part, this mode and its features are introduced.

1.3.3 Stability and transition in supercritical fluids

Ren et al. (2019a) studied the linear stability of SCF Poiseuille flow, showing that the non-ideal gas effects have strong influence on the linear stability behaviors. Ren et al. (2019b) then extended the study to boundary layer flows near the Widom line. They have shown that non-ideal effects stabilize the flow like the compressibility effects of ideal gas when the temperature profile does not cross pseudo-critical temperature. More importantly, they found that, in transcritical boundary layer, the flow is highly destabilized through a new unstable model, which is called Mode II. Figure1.7 shows the results of this study. Four temperatures and four Eckert numbers are chosen to represent different thermal and compressibility effects, and $Ec_\infty = 0.2$, $T^*_\infty = 280$ K is the only transcritical case (temperature profile cross the Widom line). This mode has a spatial growth rate one order of magnitude greater than Mode I (TS waves). They also showed that this is not related to Mack's second mode (Mack, 1984) which is attributed to acoustic effects, and suggested that Mode II has an inviscid nature. Following Ren's work, Bugeat et al. (2022) carried out an inviscid linear stability analysis on a over a flat plat boundary layer, confirming the inviscid nature of this mode. The Mach number is intentionally set to 0 for the propose of ruling out acoustic effects. In addition to recover the new mode, they also proved that this mode is related to the a minimum of kinematic viscosity at the Widom line.

While these studies focused on the linear stability analysis of Mode II, the subsequent steps towards transition to turbulence remains unexplored. The objectives of this study is closely related to this question and are presented next.

1.4 Objectives

After the discovery and investigations of Mode II, we want to further study the secondary instability of Mode II and their contribution to laminar-turbulent transition, but the breakdown process will not be in the scope. Direct numerical simulation (DNS) is used as a tool to help us understand better the nature and behaviors of different modes, and potentially resonance interactions and non-linear behaviors which are difficult to study theoretically. Linear stability analysis (LST) results are also included to be compared with DNS results. The main objective is to detect a secondary instability of Mode II in the SCF flat-plate boundary flow.

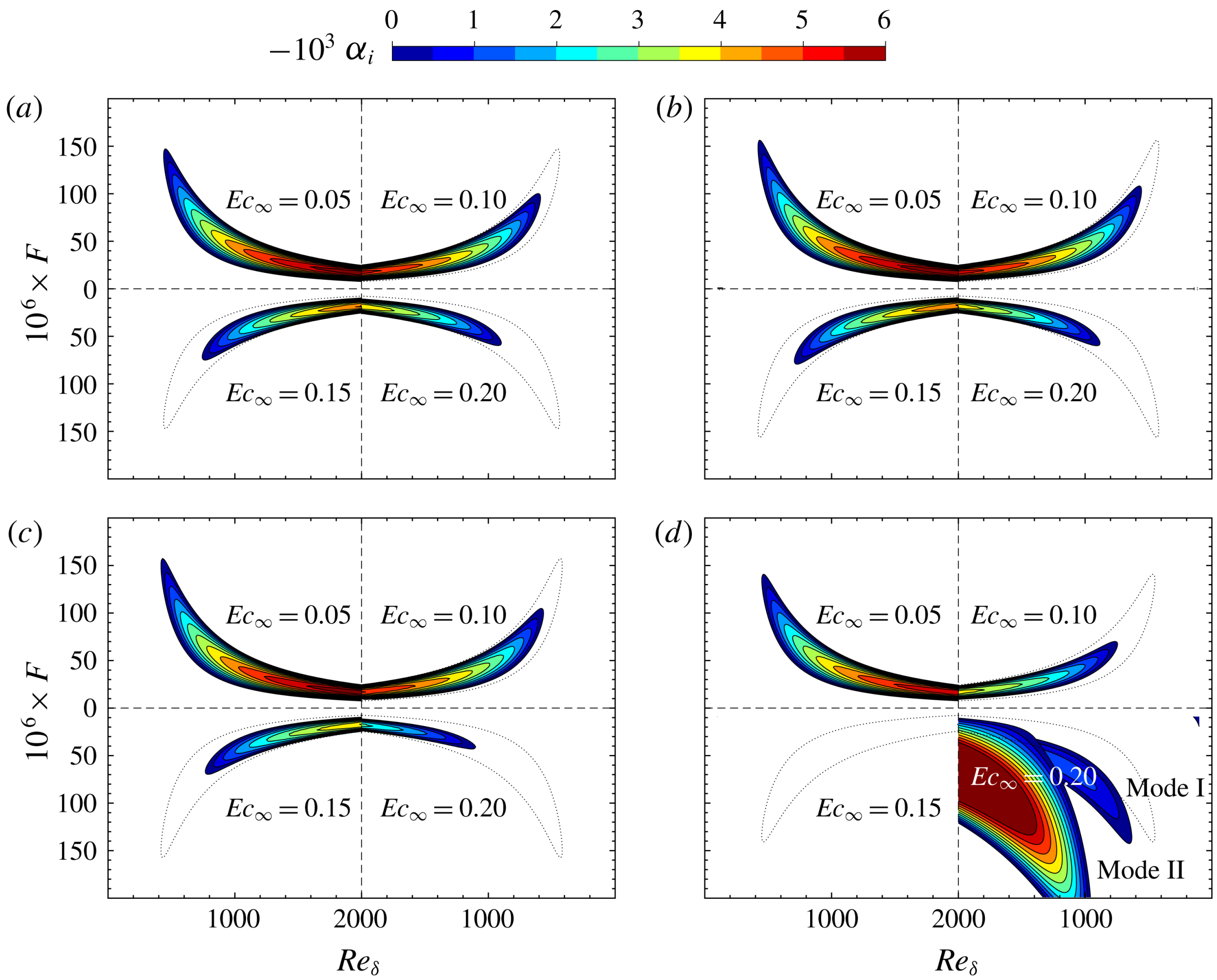


FIGURE 1.7: Growth rates of perturbations in the $F - Re_\delta$ stability diagram with subcritical free-stream temperatures: (a) $T^*_\infty = 220$ K , (b) $T^*_\infty = 240$ K , (c) $T^*_\infty = 260$ K and (d) $T^*_\infty = 280$ K. Figure from Ren et al. (2019b)

The first step consists of validating the postprocessing scripts and related data treatments, which are also used in the following parts. Ideal gas simulations are carried out for 2D and 3D case, aiming to recover linear stability theory results of primary (TS waves) and secondary (subharmonic) instabilities. Along side, study of different methods and criteria used for post-processing DNS data and identifying the growth of different modes is also effectuated. Then, DNS of SCF boundary layer are performed, intend to recover Mode II found by LST. This helps to confirm the robustness of the DNS program for SCF, whose special properties have higher requirements for stable numerical methods. Finally, subharmonic forcing is added into DNS in order to trigger secondary instability of Mode II by tuning the forcing amplitude of primary instability. When the secondary instability is found, it is discussed with the secondary instabilities of ideal gas. We also vary the spanwise wave number in order to qualitatively study its influence. With all the results of multi-frequency simulations, we discuss how the modal transition happens in a flat-plate boundary layer flow of SCFs.

Chapter 2

Theoretical and numerical approaches

This chapter introduces the analytical and numerical methods that will be deployed in this study. First, we explain how theoretical solutions are found. The governing equations are presented in part2.1, followed by a general explication of linear stability analysis and secondary instability analysis. Then, the methodology and settings for direct numerical simulation are described in part2.3, where methods for postprocessing simulation data and identifying instability evolution are also explained in detail.

2.1 Dynamical system: the governing equations

2.1.1 Navier-Stokes equations in compressible flows

The governing equations are the compressible Navier-Stokes equations. The dimensional system is made dimensionless by choosing reference values for each variable. Here we choose to non-dimensionalize variables by:

$$
\begin{aligned}
&u = \frac{u^*}{u^*_\infty}, \quad x_i = \frac{x^*_i}{l^*_0}, \quad t = \frac{t^* u^*_\infty}{l^*_0}, \quad p = \frac{p^*}{\rho^*_\infty u^{*2}_\infty}, \quad \rho = \frac{\rho^*}{\rho^*_\infty} \\
&T = \frac{T^*}{T^*_\infty}, \quad E = \frac{E^*}{u^{*2}_\infty}, \quad \mu = \frac{\mu^*}{\mu^*_\infty}, \quad \kappa = \frac{\kappa^*}{\kappa^*_\infty}
\end{aligned}
\tag{2.1}
$$

where $(\cdot)^*$ is the dimensional variable, $(\cdot)^*_\infty$ is the freestream value of a variable, and l^*_0 is a length scale. In DNS program l^*_0 is δ^*_{99}, the 99% boundary layer thickness at the inlet while in linear stability analysis, it is $\delta(x)$ the boundary layer thickness scale, with:

$$
\delta^*(x) = \sqrt{\frac{\mu^*_\infty x^*}{\rho^*_\infty U^*_\infty}} \tag{2.2}
$$

The non-dimensionlization leads to the definition of dimensionless numbers. First, the Mach number (Ma) and the Eckert number (Ec):

$$
Ma = \frac{U^*_\infty}{a^*_\infty} \qquad Ec = \frac{{U^*_\infty}^2}{c^*_{p\infty} T^*_\infty} \tag{2.3}
$$

which are related by:

$$
Ec = Ma^2 \frac{{a^*_\infty}^2}{c^*_{p\infty} T^*_\infty} \tag{2.4}
$$

where a^*_∞ the speed of sound, $c^*_{p\infty}$ the specific heat capacity and T_∞ the temperature in freestream. Note that, for ideal gas, this relation can also be further simplified by:

$$Ec = Ma^2(\gamma - 1) \tag{2.5}$$

where γ is the heat capacity ratio $\gamma = c_p^*/c_v^*$.

The Mach number measures how the characteristic velocity of flow compared to sound velocity and indicates the compressibility of the flow. The Eckert number is a ratio between the kinetic energy and enthalpy of the flow, and thus measures the viscous heating effects in the flow. If Ec is large enough, a temperature field emerges even without heat transfer (adiabatic boundary condition). When Ma and Ec are big, the compressibility is greater and acoustic effects enter in the problem. We would like to decrease these two numbers as small as possible to rule out these effects which helps the analysis. To generate a temperature gradient to create a transcritical boundary layer, a higher temperature will be imposed at the wall since Ec is small.

The Prandtl number is also an important dimensionless parameter, which is defined by

$$Pr = \frac{c_{p\infty}^* \mu_\infty^*}{\kappa_\infty^*}. \tag{2.6}$$

This parameter reflects how momentum diffusion (by viscosity) compares to heat diffusion, thus reflecting whether the boundary layer thickness defined by velocity or temperature is larger. If the viscosity is particularly large compared to the thermal diffusivity, the momentum decreasing effect of the wall is more prominent therefore the velocity boundary layer is thicker than the thermal boundary layer. Inversely, the thermal boundary layer thickness is greater.

The next dimensionless number is the Reynolds number. By convention, we use the Reynolds number based on the boundary layer thickness, Re_δ, or that based on the streamwise position, Re_x, to show the local behavior of different modes. They are defined as:

$$Re_{\delta(x)} = \frac{\rho_\infty^* U_\infty^* \delta^*(x)}{\mu_\infty^*} \qquad Re_x = \frac{\rho_\infty^* U_\infty^* x^*}{\mu_\infty^*} = Re_{\delta(x)}^2. \tag{2.7}$$

Finally we introduce the dimensionless frequency, an important dimensionless number for identifying modes with different frequencies:

$$F = \omega^* \frac{\mu^*}{\rho^* {U_\infty^*}^2} = \frac{\omega}{Re}. \tag{2.8}$$

The governing equations after non-dimensionalization are written as

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0 \tag{2.9a}$$

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j + p\delta_{ij} - \tau_{ij})}{\partial x_j} = F_i \tag{2.9b}$$

$$\frac{\partial(\rho E)}{\partial t} + \frac{\partial(\rho E u_j + p u_j + q_j - u_i \tau_{ij})}{\partial x_j} = u_j F_j \tag{2.9c}$$

where $E = e + u_j u_j/2$ the total energy, with e the internal energy, F_i the body force, τ_{ij} the viscous stress tensor and q_j the heat flux with

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k} \qquad q_j = -\kappa \frac{\partial T}{\partial x_j} \tag{2.10}$$

for real gas, the dynamic viscosity μ, the second viscosity $\lambda = -\frac{2}{3}\mu$, the thermal conductivity κ and the internal energy e are related to temperature T and the density ρ. The system of equation is closed by an equation of state (EoS).

2.1.2 Fluid model

In this study, carbon dioxide (CO_2) is chosen as the substance. Two types of equation of state (EoS) are used for modelling the thermodynamics of the fluid. The first is the ideal gas EoS

$$p = \rho R_g T \tag{2.11}$$

where R_g is the specific gas constant. This EoS well captures the $p-\rho-T$ behavior of fluids in gas state, in which the pressure is low and the temperature is high. It is not good enough to present the thermodynamic properties when the fluid is near its critical point, thus a more accurate EoS should be chosen.

Van der Waals (VdW) EoS is used for modelling real gas effects of supercritical fluids. It is written as

$$p = \frac{R_g T \rho}{1 - b\rho} - a\rho^2 \tag{2.12}$$

a is a measure of the attraction forces between molecules. b accounts for the finite volume occupied by the molecules. The constant a and b can be determined at the critical point:

$$a = \frac{27}{64}\frac{R_g^2 T_c^2}{p_c} \qquad b = \frac{R_g T_c}{8 p_c} \tag{2.13}$$

for CO_2, $p_c = 73.9$ bar and $T_c = 304.25$ K.

In addition to EoS, models for momentum and thermal transport properties are also needed for representing real gas effects. For ideal gas, Sutherland law is used, which is:

$$\frac{\mu}{\mu_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \frac{T_{ref} + S_1}{T + S_1} \tag{2.14a}$$

$$\frac{\kappa}{\kappa_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \frac{T_{ref} + S_2}{T + S_2} \tag{2.14b}$$

for CO_2:

$$T_{ref} = 273\ \text{K} \qquad \mu_{ref} = 1.37 \times 10^{-5}\ \text{Pa s} \qquad \kappa_{ref} = 0.0146\ \text{W}\ (\text{m K})^{-1}$$
$$S_1 = 222\ \text{K} \qquad S_2 = 1800\ \text{K}\ .$$

When VdW EoS is used, dynamic viscosity and thermal conductivity are modelled by the laws proposed by Jossi et al. (1962) and Stiel and Thodos (1964) , respectively.

2.2 Stability analysis

2.2.1 Linear stability analysis and method of normal modes

The first step of linear stability analysis is to decompose the flow field into the base flow and a perturbation:

$$Q = \bar{Q} + q \tag{2.15}$$

where Q represents a variable among density ρ, velocity field u_i, temperature T, pressure p, energy E, and coefficients μ, κ. $\bar{Q}$ is the base flow and also a solution of equations (2.9) and EoS. Injecting each variable in the form of (2.15) into the system of equation, subtracting the parts already satisfied by the base flow and neglecting all second order terms of perturbation by assuming the magnitudes of perturbations are small, we get a linearized system:

$$\frac{\partial \mathbf{q}}{\partial t} = \mathcal{V}(\bar{\mathcal{Q}})\mathbf{q} \tag{2.16}$$

where $\mathbf{q}$ is a vector of perturbation variables, and $\mathcal{V}$ is a matrix as a function of the base flow.

Assuming the base flow is homogeneous in the spanwise direction and also in the streamwise direction because the characteristic length of boundary layer evolution is much greater than the wave length in the streamwise direction, under linear stability assumption, the modes of perturbations can be expressed as

$$\begin{aligned} q(x,y,z,t) &= \frac{1}{2}\left[\hat{q}(y)\exp\Big(i(\alpha x + \beta z - \omega t)\Big) + \text{c.c.}\right] \\ &= \Re\left[\hat{q}(y)\exp\Big(i(\alpha x + \beta z - \omega t)\Big)\right] \end{aligned} \tag{2.17}$$

where c.c. stands for complex conjugate, and q is the quantity of interest. In this study, we consider that instabilities grow spatially in the streamwise direction but not temporally, so that β and ω are real numbers, whereas α is a complex number: $\alpha = \alpha_r + i\alpha_i$. The modes can then be written as

$$q(x,y,z,t) = \exp\big(-\alpha_i x\big)\cdot \Re\left[\hat{q}(y)\exp\Big(i(\alpha_r x + \beta z - \omega t)\Big)\right] \tag{2.18}$$

where α_i is the growth rate.

Injecting expression (2.17) into the system of equation (2.16) gives

$$\mathcal{M}(\alpha,\beta,\omega)\cdot\hat{\mathbf{q}} = 0 \tag{2.19}$$

where $\mathcal{M}$ is a matrix, and the system can be solved as an eigenvalue problem. The expression of exact terms in equation (2.16) is shown in detail in the work of Ren et al. (2019a).

2.2.2 Secondary instability analysis

It is mentioned in sections 1.1 and 1.2 that secondary instability appears on a new periodic base flow composed of the original based flow and the primary mode with finite amplitude, and it may be an important precursor of transition to turbulence (modal route). The secondary instability analysis is similar to the linear stability analysis but makes use of the fact that the base flow is periodic which allows us to use Floquet analysis (Herbert et al., 1987) to solve the system of equations. Although these equations are not actually solved in the present work, they are introduced in this section since the identification of secondary instabilities via DNS calculations will be based on this framework.

Assuming that the flow field components of primary instability of T-S waves are written as $q^{2D}(x', y)$, where c_r is the phase velocity and $x' = x - c_r t$. By definition, the original coordinate is transformed into one that travels with the waves. Then the velocity field of the new base flow can be written as

$$U_i = [U(y) - c_r]\delta_{i1} + A[u^{2D}(x',y)\delta_{i1} + v^{2D}(x',y)\delta_{i2}] \tag{2.20}$$

where A is a measure of the amplitude of the primary waves. In the new frame of reference, the base flow is spatially periodic, verifying

$$U_i(x',y) = U_i(x' + \lambda_x, y) \tag{2.21}$$

where λ_x is the wavelength of the TS waves.

Following the same procedures of linear stability analysis, we can derive the system of equation for secondary instability components in the coordinates (x', y, z). This is a system of differential equation with periodic coefficients in x'. Floquet theory then indicates that the solutions to these equations take the form:

$$q(x', y, z, t) = \hat{q}(x', y)e^{\gamma x'}e^{\sigma t}e^{i\beta z} \qquad \gamma, \sigma \in \mathbb{C}, \quad \beta \in \mathbb{R} \tag{2.22}$$

where $\hat{q}(x', y)$ is a periodic function in x' with period λ_x and γ is the Floquet parameter. Applyin Fourier transformation to $\hat{q}(x', y)$, the general form of the solution can be written as:

$$q(x', y, z, t) = e^{\gamma x'}e^{\sigma t}e^{i\beta z}\Sigma_m \hat{q}_m(y)e^{im\alpha x'} \tag{2.23}$$

Assuming $\gamma = \gamma_r + i\gamma_i$ and $\sigma = \sigma_r + i\sigma_i$, similar to the linear stability analysis, only two among γ_r, γ_i, σ_r, σ_i are determined by the eigenvalue problem associated to secondary instability and the other two are chosen. Because of the periodic nature of complex number expressed in Euler's formula, γ and $\gamma \pm in\alpha$ (with $n \in \mathbb{N}$) yield identical modes, therefore it is sufficient to study $-\alpha/2 < \gamma_i \leqslant \alpha/2$. By introducing $\epsilon = \gamma_i/\alpha$, three modes can be distinguished by ϵ:

Fundamental modes with $\epsilon = 0$:

$$q(x', y, z, t) = e^{\gamma_r x'}e^{\sigma t}e^{i\beta z}\Sigma_m \hat{q}_m(y)e^{im\alpha x'} \tag{2.24}$$

Subharmonic modes with $\epsilon = \frac{1}{2}$:

$$q(x', y, z, t) = e^{\gamma_r x'}e^{\sigma t}e^{i\beta z}\Sigma_m \hat{q}_m(y)e^{i(m+\frac{1}{2})\alpha x'} \tag{2.25}$$

Detuned modes with $0 < \epsilon < \frac{1}{2}$:

$$q(x', y, z, t) = e^{\gamma_r x'}e^{\sigma t}e^{i\beta z}\Sigma_m \hat{q}_m(y)e^{i(m+\epsilon)\alpha x'} \tag{2.26}$$

With these definitions, we can observe that in coordinate (x', y, z), the streamwise wave number of the fundamental mode is the same as TS waves while that of the subharmonic mode is half of the TS waves. When spatial growth in laboratory coordinate is considered, $\sigma_r = \gamma_r c_r$. γ_r indicates the spatial growth rate, and the frequency is given by $\sigma_i - \gamma_i c_r$. These expressions also explain various observations, recalling Figure1.4. The fundamental modes are doubly periodic with wavelengths λ_x and λ_z, accounts for the aligned patterns. The subharmonic modes are doubly periodic with wavelengths $2\lambda_x$ and λ_z, and invariant under the translation $(x, z) \rightarrow (x + \lambda_x, z + \lambda_z/2)$, which corresponds to the observed staggered patterns.

2.3 Numerical methodology

Apart from stability analysis, direct numerical simulation (DNS) is also a generally acknowledged way to study instability behaviors and laminar-turbulent transition. Compared to LST, it solves directly the governing equations so the results are closer to what happens in the physical world. Also, LST is based on numerous simplifications, if DNS can recover LST results, the simplifications made by LST are validated, therefore prove which effects are negligible. Moreover, when the amplitude of perturbation becomes large and non-linear effects enter the problem, LST predictions no longer holds. Therefore, DNS is a more adapted method to study non-linear effects which may probably appear in this study. However, DNS is more computationally expensive as high spatial and temporal resolution are required to get accurate results, as a consequence, the number of cases can be studied is limited by the available computational time and power.

For the purpose of generating and identifying primary and secondary instabilities, data are produced with an in-house DNS program written in `FORTRAN` and then post-processed in `MATLAB`. The in-house DNS codes used in this study employs finite difference scheme for spatial discretization and explicit third-order Runge-Kutta scheme for the temporal integration. The convective terms are evaluated by *generalized conservative approximations* (Pirozzoli, 2010) to guarantee numerical stability and energy conservation.

2.3.1 Computational setup

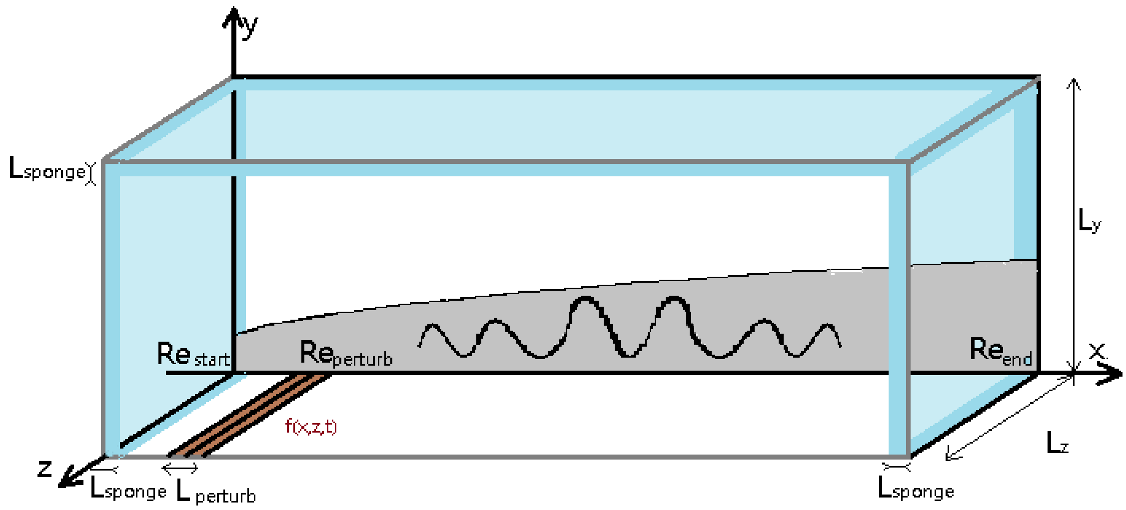


FIGURE 2.1: Computational domain

Figure2.1 illustrates how the computational domain is defined in DNS program with different parameters. The dimension of the domain is defined by the height of the domain L_y, the spanwise length of the domain L_z, and the Reynolds number based on $\delta(x)$ Re_{Start} and Re_{End} that specifies the start and end streamwise positions.

Regarding to the mesh, n_x, n_y and n_z define respectively the number of grid points in streamwise, wall normal and spanwise direction. The grid is equally spaced in the streamwise and spanwise direction and refined at the wall in the wall normal direction using a function with hyperbolic tangent

$$x(i) = L_y * \left[C \cdot \eta + (1 - C) \cdot \left(1 + \frac{tanh(S_f(\eta - 1)/2)}{tanh(0.5 S_f)} \right) \right] \tag{2.27}$$

where

$$\eta(i) = \frac{i-1}{N-1}, \qquad C = \frac{0.8}{Re_\tau} \frac{N-1}{L_y}, \qquad Re_\tau = \delta(x_{start}) \sqrt{\frac{\rho_{wall}}{\mu_{wall}} \frac{dU}{dy}}, \tag{2.28}$$

such that the mesh can be refined by increasing the stretching factor S_f.

Non-reflecting boundary conditions are used to avoid non-physical reflecting waves. Their implementation is based on inviscid characteristic wave analysis (Poinsot and Lelef, 1992), and are implemented at the inlet, outlet, top and bottom of the domain. In addition to that, numerical sponge is added at the inlet, outlet and top of the domain in order to suppress the non-physical oscillations caused by waves reflected from the boundaries. $L_{sp,in}$, $L_{sp,out}$ and $L_{sp,fs}$ marks respectively the sponge length at inlet, outlet and on the top, and they are adjusted to each simulation by trial and error such that the result doesn't depend on sponge length. Periodic boundary conditions are used for the left and right side of the domain.

For the boundary condition at the wall, we choose either adiabatic or isothermal non-reflecting boundary condition. The forcing is introduced in form of blowing suction at the wall. It is defined as an unsteady boundary condition, characterised by its position, length, amplitude and frequency. Formally, the wall normal velocity at the wall is imposed by a function $f(x, z, t)$ defined as

$$f(x, z, t) = A_1 f_x(x) \sin(\omega_1 t) + A_2 f_x(x) g_z(z) \sin(\omega_2 t) \tag{2.29}$$

where:

$$g_z(z) = \cos(2\pi z/\lambda_0) \tag{2.30}$$

$$|f_x(x)| = 15.11875\xi^5 - 35.4375\xi^4 + 20.25\xi^3 \tag{2.31}$$

$$\xi = \begin{cases} \frac{x-x_1}{x_m-x1} & \text{for} \quad x_1 \leq x \leq x_m \\ \frac{x-x_2}{x_m-x2} & \text{for} \quad x_m \leq x \leq x_2 \end{cases} \tag{2.32}$$

This choice is chosen and validated by several studies of instability development in a boundary layer, for example, the work of Fasel and Konzelmann (1990) and Sayadi et al. (2013). This equation defines a forcing between position x_1 and x_2. The first term of equation2.29 intends to trigger 2D primary instabilities with angular frequency ω_1, the second term stands for 3D forcing with angular frequency ω_2 and spanwise wave length λ_0. A_1 and A_2 are respectively the forcing amplitudes for modes with frequency ω_1 and ω_2.

2.3.2 Identification of instabilities

After flow motion reaches to periodic state, which means the amplitudes of fluctuations are constant in time, DNS data are saved with a certain sampling frequency such that the number of samples in one period is an integer. This is for the sake of carrying out fast Fourier transformation (FFT) in time without introducing any sampling error. More precisely, if DNS data over N period with m samples in one period are to be analyzed with FFT, $N \times m - 1$ samples should be used.

We would like to identify the mean flow value and fluctuation value from DNS data. The mean flow $\bar{Q}$ can be derived either by time averaging or choosing the $F = 0$ value from FFT analysis. The fluctuations q are seen as a combination of normal modes, expressed in equation (2.18). In the following analysis, only one mode with certain frequency ω_n and spanwise wave number β_n is under consideration, and we would like to derive the related spatial growth rate $-\alpha_{i,n}$.

In numerical data analysis, the growth rate $-\alpha_i(x)$ of a quantity $A(x) = A_0 \exp\left[-\alpha_i(x) \cdot x\right]$ at a chosen position x_0 can be calculated by:

$$-\alpha_i = \frac{1}{A}\frac{dA}{dx}\bigg|_{x0} \tag{2.33}$$

However, the mode does not only depend on x. To find the quantity $A(x)$ and extract the growth rate from DNS data, one should choose a quantity of interest and find a method to calculate $A(x)$ from $q(x, y, z, t)$.

FFT method is applied in time and in spanwise direction to q, and its absolute value at frequency ω_n and spanwise wave number β_n is called *Fourier amplitude*. With the Fourier amplitude of each variable, we would like to compute the quantity of interest. An energy related quantity or one of the variables itself can be used, and some possible choices are shown in table 2.1.

Distribution of the kinetic energy:	$E_k = \bar{\rho}(\lvert u'\rvert^2 + \lvert v'\rvert^2)$
Distribution of the kinetic energy in incompressible case:	$e_k = \lvert u'\rvert^2 + \lvert v'\rvert^2$
Distribution of u' component contribution of the kinetic energy:	$e_u = \lvert u'\rvert^2$
Streamwise fluctuation itself:	u'

TABLE 2.1: Considered quantity for representing instability intensity. $|u'|$ and |v'| are the Fourier amplitude of u' and v', $\bar{\rho}$ is the mean value of density

However, these quantities still depend on wall normal position. Table 2.2 shows four possible methods to calculate the wall normal position independent value. These methods are chosen because of the hydrodynamic and instability characteristics of boundary layer flows. The first method goes with the energy related quantity in table 2.1, representing a streamwise distribution of fluctuation energy. According to LST prediction, the profile of $|u'|$ is a function of $y/\sqrt{x}$ and it has a maximum near the wall. The second and fourth methods are come up as a consequence. And the third method is brought forward by the fact that boundary layer is very thin and its thickness grow slowly, so measuring at a fixed y position is a relatively simple approach which can give representative results. When implemented numerically, the second and

Integration:	$\int_0^{L_y}(\cdot)dy$
Maximum value in the normal direction:	$\max_y(\cdot)$
Value of the quantity at a fixed y:	$(\cdot)(x, y = y_{fixed})$
Value of the quantity following $y = l(x) \propto \sqrt{x}$:	$(\cdot)(x, y = l(x))$

TABLE 2.2: Methods for treating wall normal position dependency

third one are straight forward. For the first one, midpoint numerical integration is used, which means

$$\int_y e(x,y)dy = \sum_{i=0}^{N-1}(\frac{e_i + e_{i+1}}{2} \times \Delta y). \tag{2.34}$$

For the fourth method, the value of a quantity $e(x, y)$ at position $(x, y = A\sqrt{x})$ is determined by linear interpolation. To be more specific, suppose $y_i = A\sqrt{x_i}$ is situated between y_1 and y_2 ($y_1 < A\sqrt{x_i} < y_2$) with corresponding values $e_1 = e(x_i, y_1)$ and $e_2 = e(x_i, y_2)$, then

$$e(x_i, A\sqrt{x_i}) = \frac{y_i - y_2}{y_1 - y_2}e_1 + \frac{y_i - y_1}{y_2 - y_1}e_2 \tag{2.35}$$

Considering the physical meaning and the simplicity of different criteria and methods, certain combinations are preferred than others. For example, in experiments, measuring values at fixed y position is preferred compared to integration-based methods. In numerical studies, integrating along wall normal direction is not as much complicated as in experiments, and can make most use of simulation data, therefore is possibly a better choice. Another reason for choosing a certain quantity or method is because it was used and validated in previous studies. While more criteria are implemented and tested in the validation step with the help of 2D ideal gas simulations, only three of them are finally chosen to be used throughout the whole study, shown in table 2.3.

(c)	$\int_0^{L_y} e_u dy = \int_0^{L_y} \lvert u'\rvert^2 dy$
(d)	$\lvert u'\rvert$ at fixed wall normal position: $\lvert u'\rvert\big\rvert_{y_{fix}}$
(g)	$\max_y \lvert u'\rvert$

TABLE 2.3: The chosen criteria validated with 2D simulation. $|u'|$ is the Fourier amplitude of u'

By convention, the dimensionless growth rate use the local boundary layer thickness $\delta^*(x)$ as the length scale, while the length scale used in DNS simulations is δ_{99}^* at the inlet. To ensure consistency between DNS and LST results, the growth rate calculated via DNS data is actually

$$-\alpha_i = \delta(x)\frac{1}{A}\frac{dA}{dx^*} = \frac{\delta(x)}{\delta_{99}}\frac{1}{A}\frac{dA}{dx} = \frac{Re_{\delta(x)}}{Re_{\delta_{99}}}\frac{1}{A}\frac{dA}{dx} \tag{2.36}$$

It is also possible to compare directly the amplitude $A(x)$ of a mode without calculating growth rate. In this case, we can derive the characteristic value $A(x)$ from theoretical growth rate by using the following expression:

$$A(x) = \frac{Re_{\delta_{99}}}{Re_{\delta(x)}} \cdot A_0 \cdot \exp\left[-\int_{x_0}^{x} \alpha_i(\xi)d\xi\right] \tag{2.37}$$

We note that a coefficient $\frac{1}{2}$ or 2 is added in these expressions when energy related criteria (the ones with $|\cdot|^2$) are used. In this case, the previous expressions then become

$$-\alpha_i = \frac{1}{2}\frac{Re_{\delta(x)}}{Re_{\delta_{99}}}\frac{1}{A}\frac{dA}{dx} \qquad A(x) = \frac{Re_{\delta_{99}}}{Re_{\delta(x)}} \cdot A_0 \cdot \exp\left[-\int_{x_0}^{x} 2\alpha_i(\xi)d\xi\right] \tag{2.38}$$

Finally, the streamwise wave number α_r and phase velocity c_r of the instability can also be derived by DNS data with

$$\alpha_r(x) = \frac{\partial\phi}{\partial x} \tag{2.39}$$

$$c_r(x) = \omega/\alpha_r(x) = Re_{\delta_{99}} \cdot F/\alpha_r(x) \tag{2.40}$$

where $\phi = arg[FFT(q)]$, the phase angle of the perturbation, and F the dimensionless frequency in expression (2.8).

Chapter 3

Ideal gas simulations

With the objective of analysing the secondary instabilities in supercritical fluid via DNS, the first step is to validate the postprocessing methods and the associated `Matlab` script with ideal gas simulations by recovering LST prediction of the eigenmodes. These first simulations also help us to study the mesh convergence, to understand the influence of numerical parameters, and to discuss different criteria for identifying instability behaviors. This chapter begins with a study of different criteria, followed by discussions of numerical parameters, and ended with a presentation of 3D simulation results recovering the H-type secondary instability growth.

3.1 Identification of the primary instability

To begin with, 2D DNS with one forcing frequency and small forcing amplitude is set up to recover the primary instability, i.e. TS waves, introduced in chapter 1.2. As the physical parameters and boundary conditions in DNS are set to match the ones in LST analysis, recovering LST prediction by DNS results validates the post-processing scripts. Additionally, the first DNS data also help us to study different criteria and determine which ones to use.

The parameters used as reference case in 2D ideal gas simulations are listed in table A.1 by the case name **2D IG REF**. The validation of domain height and number of grid points are shown in appendix B. This section begins with a study of different criteria, followed by discussion about the influence of parameters.

3.1.1 Study of different criteria

In general, we would like to find a criterion that gives the expected results with less data. Additionally, it is better that the criterion depends on fewer coefficients and is easy to implement. Apart from the validated criteria mentioned in section 2.3.2, table 3.1 shows all the criteria to be discussed in this part. More criteria were brought forward as combinations of different quantity and methods, but some of them are rejected for a lack of physical meaning or for simplicity reasons. This caused the numbering in table 3.1.

(c)	$\int_0^{L_y} e_u dy = \int_0^{L_y} \lvert u'\rvert^2 dy$
(d)	$\lvert u'\rvert$ at fixed wall normal position (different y_{fix} are tested)
(e)	$\lvert u'\rvert$ following $y = l(x) \propto \sqrt{x}$ (different coefficients before $\sqrt{x}$ are tested)
(g)	$\max_y \lvert u'\rvert$

TABLE 3.1: Criteria studied with 2D ideal gas reference case. $|u'|$ is the Fourier amplitude of u'

Because we think the criterion (c) has the best physical meaning, it is chosen as a reference apart from LST prediction to be compared with the other three criteria. Noting that $|u'|$ has a power of two in criterion (c), to match its value, the amplitude curve $A(x)$ of other criteria are actually $A^2(x)$. To begin with, figure 3.1 shows the results of criterion (d) using different wall normal positions. Apparently, the amplitudes and the related growth rates depend on the chosen wall normal position, especially in upstream. Apart from that, the growth rate becomes significantly different when wall normal position is greater than 1.0. This is expected as dimensionless value 1 equals to the 99% boundary layer thickness at the inlet, so when y_{fix} is too big, the values are not measured within boundary layer. Although the instability growth depends on the chosen wall normal position, the differences are not important when a decent choice of y_{fix} is made. With an appropriate y_{fix} value, the curves are in good agreement with LST prediction. This is indeed a method that has been used in several numerical and experimental investigations in the past for its easy implementation and little storage requirement. With our observation, it is again confirmed to be a validated choice.

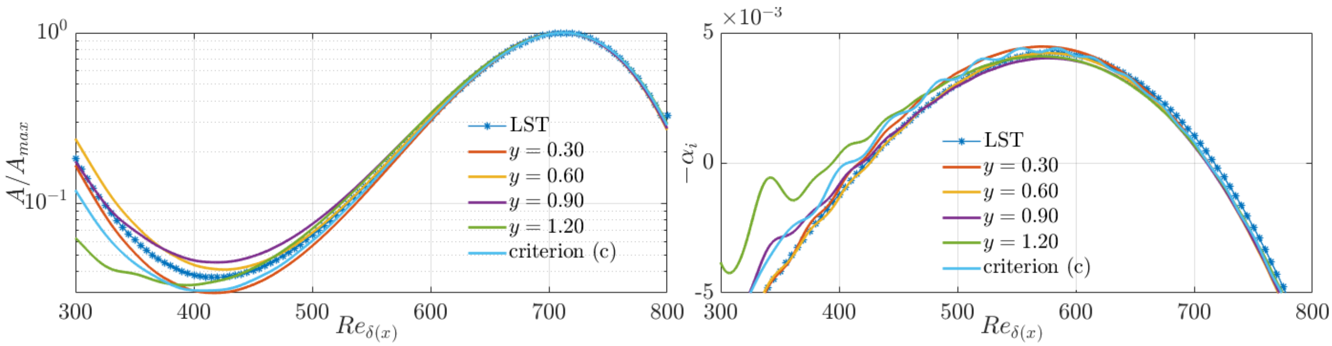


FIGURE 3.1: Amplitude (left) and related growth rate (right) for criteria (d) in table 3.1 with different y_{fix}, compared with integrated method (c) and LST

Method (e) in table 3.1 is also related to the wall normal position. The values are measured with $y = l(x) \propto \sqrt{x}$ and different coefficients are chosen in front of $\sqrt{x}$. Figure 3.2 shows the results. When

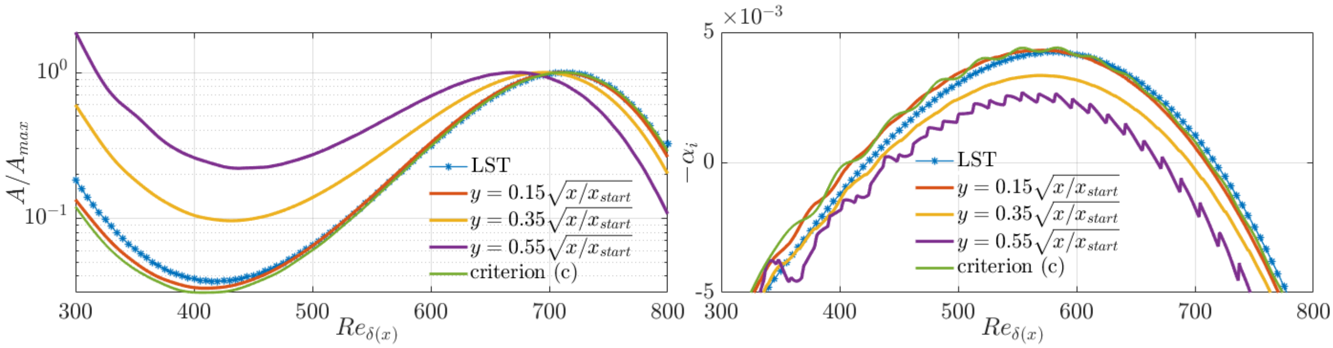


FIGURE 3.2: Amplitude (left) and related growth rate (right) for criteria (e) in table 3.1 with different coefficients in front of $\sqrt{x}$, compared with integrated method (c) and LST

following $y = 0.55\sqrt{x/x_{start}}$, the growth rate curve has jagged patterns and the amplitude is the most different among all results. However, $y = 0.6\sqrt{x/x_{start}}$ is well within the boundary layer, so this is not caused by the same reason as the large y_{fix} value. A possible explanation is that higher order interpolation is needed when the values are measured further away from wall, as the mesh becomes coarser at higher position. Similar to criterion (d), criterion (e) depends on the coefficient in front of $\sqrt{x}$, but when an appropriate choice is made, the amplitude and related growth rate are coherent with that of the integrated method and LST. In spite of this, compared to the fixed y method, this method is more restricted in the choice of coefficient and do not have the advantages of easy implementation or saving storage. Thus, criterion (e) will not be used in the following study.

Finally, we show the results of the last criterion in table 3.1 in figure 3.3 together with integrated method (c) and LST results. The values found by this criterion match best with LST results among all, and the

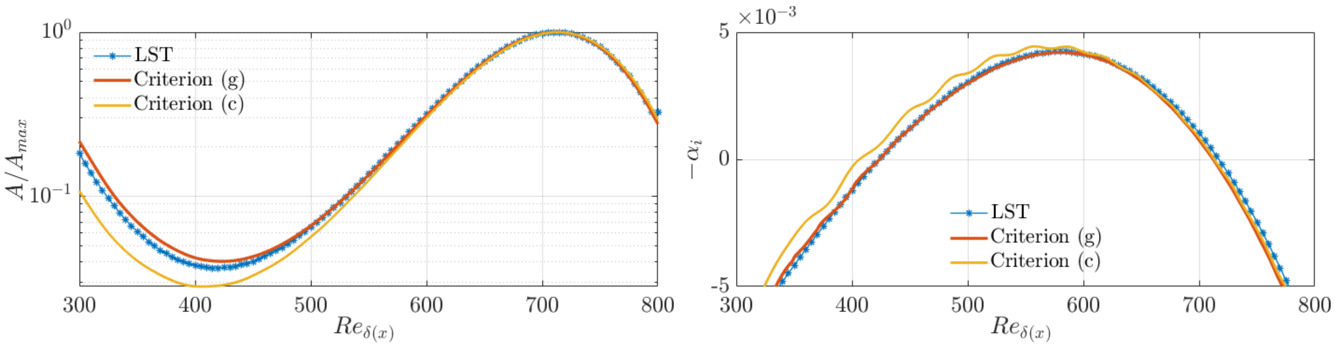


FIGURE 3.3: Amplitude (left) and related growth rate (right) for criteria (g) in table 3.1 compared with integrated method (c) and LST.

upstream oscillations in growth rate also seem to be less significant. This validates this criterion, and we would like to discuss whether it is indeed the best criterion with DNS cases that use different physical parameters.

3.1.2 Influence of numerical and physical parameters

To discuss how numerical and physical parameters influence the results, this part begins with a study of data sampling, demonstrating how the number of sample and period are chosen for post-processing with FFT method. Then results of simulations with a different frequency and with a different Mach number are presented to further verify the chosen criteria in other physical cases,

Sampling parameters: required period and sampling frequency

Here we demonstrate how many periods and how many samples per period need to be used for FFT method to get the expected result. This discussion uses the criterion (c) but the results are general for all criteria. The sampling frequency is reflected by the number of samples in one period. Figure 3.4 shows the results found by different sampling frequencies represented by the number of samples in one period. Apparently,

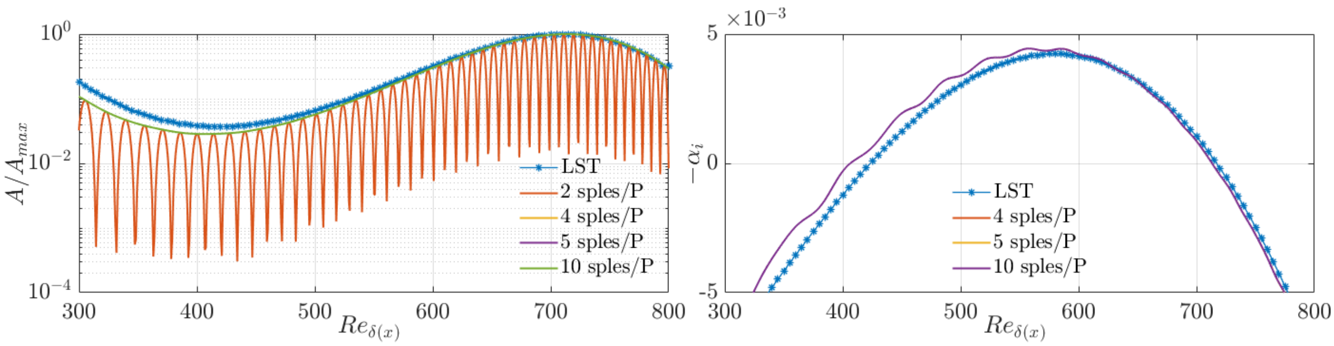


FIGURE 3.4: Amplitude (left) and related growth rate (right) using criteria (c) with 2, 4, 5 and 10 samples in period over 1 period

2 samples in one period is not enough and when the number of samples in one period is equal or greater than 4, the curves are smooth and results overlap. For the number of period, the results of 1, 2 or 3 periods with 4 samples in each period superimpose perfectly, the plots are not showing here to avoid redundancy. Consequently, only 1 period is needed for deriving the numerical result when enough number of samples is used in 1 period.

To conclude, when only one harmonic with frequency f_i is presented in the fluctuation, using $4f_i$ as the sampling frequency over only one period can give the expected growth curves. This observation is consistent with the sampling theorem (Shannon, 1949). When other harmonics are present in the fluctuation, the sampling frequency should be chosen with the highest harmonic.

Application to lower frequency

Although LST prediction and DNS results are comparable enough to validate the post-processing methods and scripts, differences still exit. As mentioned in section 1.2, the non-parallel effects are less important for lower fluctuation frequencies, and the growth rate depends less on the way of instability identification. For this reason, a smaller frequency is used in the hope of getting a better match between LST prediction and DNS results. As it is shown in figure 1.3, the unstable range becomes larger for lower frequencies. Therefore, the computational domain is redefined by the parameters in table A.1 under the case name **2D IG F33**. The results are shown in figure 3.5. Compared to the previous case, the consistency between different criteria and LST prediction indeed becomes better, thus again confirms the listed criteria.

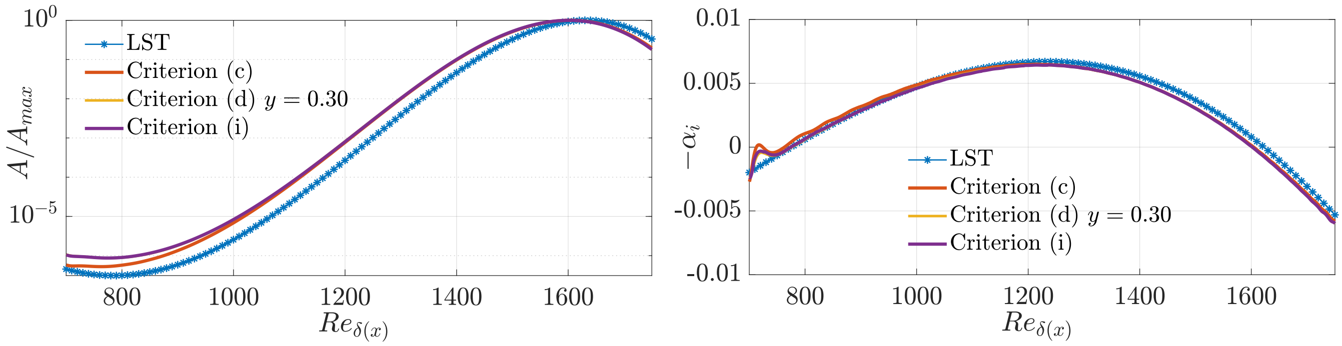


FIGURE 3.5: Amplitude (left) and related growth rate (right) using criteria listed in table 2.3 with $F = 33 \times 10^{-6}$

Application to another Mach number

The DNS program used in this study is a compressible solver, so the Mach number cannot be set to 0. To make DNS results comparable with that of incompressible LST, the Mach number used in DNS simulation needs be small. However, when decreasing Mach number, a smaller time step should be used to satisfy the same CFL criterion, because the sound velocity becomes smaller. By decreasing $Ma = 0.35$ to $Ma = 0.1$, the required number of time steps nearly doubles, so as to the computational time. Nevertheless, results of $Ma = 0.35$ don't deviate much from that of LST, so it is chosen in the previous case. In this part, a smaller Mach number $Ma = 0.1$ is used and the results are compared to that of Mach number $Ma = 0.35$ with criteria (c) and (g), shown in figure3.6. In general, whatever the criterion, the growth rate curve shifted to right when a smaller Mach number is used. This indicates that decreasing Mach number can delay instability growth. In the previous simulation with $Ma = 0.35$, growth rate curve found by criterion (g) corresponds best to LST curve. However, for $Ma = 0.1$, growth curves of criterion (c) approaches more to the LST curves. This shows that it is hard to conclude which criterion is the best. This discussion also helps us to understand the compatibility between incompressible LST prediction and results given by a compressible solver with small Mach number.

3.1.3 Discussion of the upstream oscillations

In the previous parts, oscillations can be observed in upstream in all plots of growth rate, whereas the amplitude plots are very smooth. The oscillations that we observe in upstream is different from the inappropriate value if y_{fix} in criterion (d) or bad interpolation in criterion (e). One possible origin is the

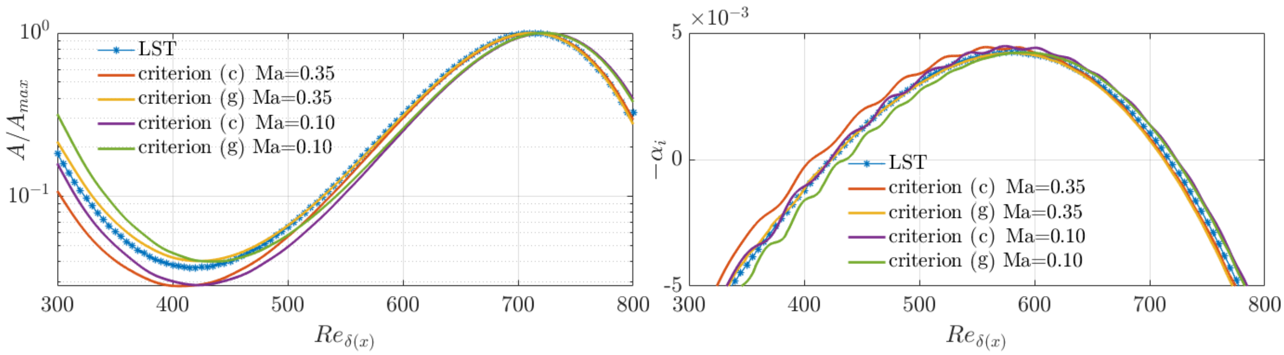


FIGURE 3.6: Amplitude (left) and related growth rate (right) of different Mach number for criteria (g), (c) and LST.

presence of another mode triggered by the blowing/suction forcing. Although only one forcing frequency is used, the blowing/suction forcing is not in the form of TS waves, it is applied as a boundary condition at the non-slippery wall. We would like to discuss these oscillations in this part. First, we show an assumption about how growth rate calculation amplifies small errors. Then, simulation results with different form and length of forcing are compared and discussed.

How growth rate calculation amplifies numerical fluctuation: a possible explanation

Apart from the oscillations in upstream, other oscillations that are imperceptible in the amplitude curve but present in the growth rate curve also have been observed during this study, showing growth rate calculation can amplify tiny numerical fluctuations. In fact, it is the errors or fluctuations with small characteristic wave length that can be amplified. Here we propose a possible explanation how it happens.

The growth rate is calculated by 2.36. Now assuming there is a tiny fluctuation $b(x)$ hiding in the characteristic value $A(x)$ so that the calculated growth rate actually is :

$$-\tilde{\alpha}_i = \frac{Re_{\delta(x)}}{Re_{\delta_{99}}} \frac{1}{A(x)+b(x)} \frac{d\left[A(x)+b(x)\right]}{dx} \qquad \text{with: } |b(x)| \ll |A(x)| \tag{3.1}$$

Considering b is much smaller than A, we apply Taylor expansion around A to the term $\frac{1}{A(x)+b(x)}$:

$$\begin{aligned} -\tilde{\alpha}_i &= \frac{Re_{\delta(x)}}{Re_{\delta_{99}}} \left(\frac{1}{A(x)} - \frac{b(x)}{A^2(x)} + \circ(b(x))\right) \left(\frac{dA(x)}{dx} + \frac{db(x)}{dx}\right) \\ &\approx \frac{Re_{\delta(x)}}{Re_{\delta_{99}}} \frac{1}{A(x)} \frac{dA(x)}{dx} \left(1 - \cancel{\frac{b(x)}{A(x)}}\right) \left(1 + \frac{b'(x)}{A'(x)}\right) \\ &\approx -\alpha_i \left(1 + \frac{b'(x)}{A'(x)}\right) \end{aligned} \tag{3.2}$$

When the variation length scale of $b(x)$ is much smaller than that of $A(x)$, the term $\frac{b'(x)}{A'(x)}$ may no longer be insignificant and appears in the growth rate curve.

Influence of the forcing function

Assuming that the oscillations upstream come from other modes, we want to decrease it by change the from of forcing function to be more similar to T-S waves. Three other forcing functions are used to compare with the original one. The criterion (g) $\max_y |u'(x,y)|$ is used to calculate the growth rate, because it gives results that have the best agreement with the theoretical ones, so the quantity $\alpha_i - \alpha_{theory}$ is centered near zero and shows more clearly how oscillations change. The interval over which the forcing is applied is fixed, with $Re_{Perturb} = 250$ and $L_{Perturb} = 33$, approximately one wavelength of the T-S waves. The original forcing function is replaced by three other functions separately. The first two consist of replacing equation2.32 by sinus or Gaussian function. In the first plot of figure3.7, the original equation2.32 and the replacement functions are shown with a transformation from x to $\chi = \frac{2x-(x_1+x_2)}{x_2-x_1}$, where x_1 and x_2 are the start and end positions of the forcing. The third option is wave forcing, which means replacing $f_x(x)\sin(\omega t)$ in equation2.29 altogether with $\sin(\pi\chi - \omega t)$, and we specially point out that violent discontinuity can arise by using this form, because wall velocity is set to zero outside the interval where the forcing is applied.

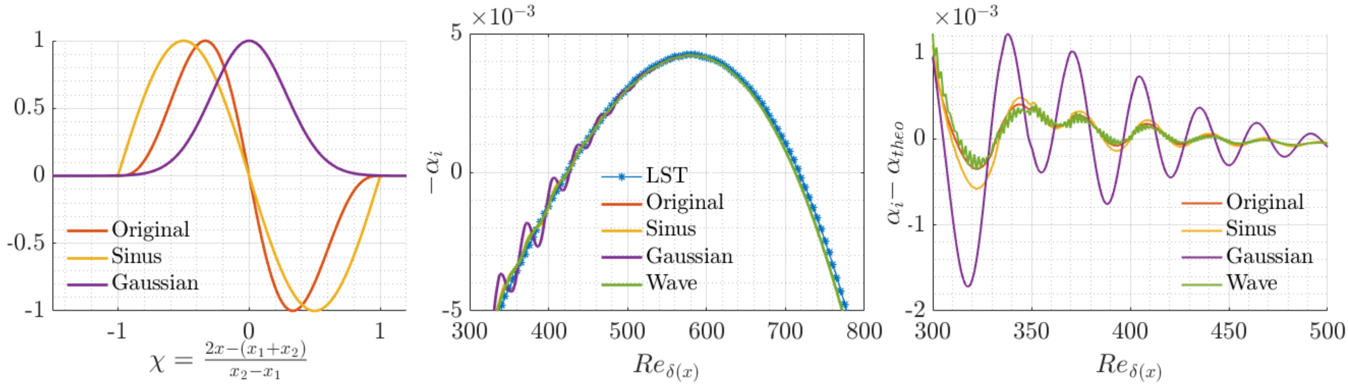


FIGURE 3.7: Forcing function (left), related growth rate (centre) and difference between each growth rate obtained by DNS and LST (right). Sinus: $f(x) = \sin(\pi\chi)$; Gaussian: $f(x) = \exp[(\frac{\chi}{0.4})^2]$

The growth rate found by different forcing functions and the differences from the theoretical one are shown in the last two plots of figure3.7. Apart from the Gaussian forcing, the other results are quite similar. This may be explained by the fact that Gaussian function is the most different from a standard sinus function and can trigger other modes with bigger amplitude. Another notice is that the wave function gives tiny oscillations and it might be caused by the discontinuity of this forcing function. Apart from the tiny oscillations, results of wave function give less oscillations than the sinus function. This support the assumption that upstream oscillations is caused by the difference between blowing/suction forcing and TS waves. Finally, we conclude that the original forcing function is a good choice. When it is used, the amplitude of upstream oscillations is smaller and there is no tiny oscillations because this function is continuous until second order at the end points.

Influence of the forcing length

To make the forcing function resembles to T-S waves, we can also set the forcing length exactly as the wave length of the T-S waves. That is why the forcing length 33 is chosen in the previous part. However it doesn't seems to decrease the oscillations. The influence of the forcing length is studied more in detail by using different forcing length. Fixing $Re_{perturb} = 250$ and use the original forcing function, growth rates of $L_{perturb} = 5, 10, 20$ and 33 are shown in figure3.8. In the first figure, we can see that the difference between each result is small. In the other two figures, the DNS results are subtracted by the theoretical value, so the differences can be observed more clearly. We notice that oscillation amplitude decreases as forcing length decreases. However, when $L_{perturb} = 5$, tiny oscillations appear. In this case, the forcing is represented by only 5 grid points so the continuity of the forcing is bad, then explains this observation. What the results show is different from our assumption but we can say that only by setting the forcing length as the wave length is not enough to trigger "clean" T-S waves and larger forcing length may trigger other modes with larger amplitude, but this influence is not very strong.

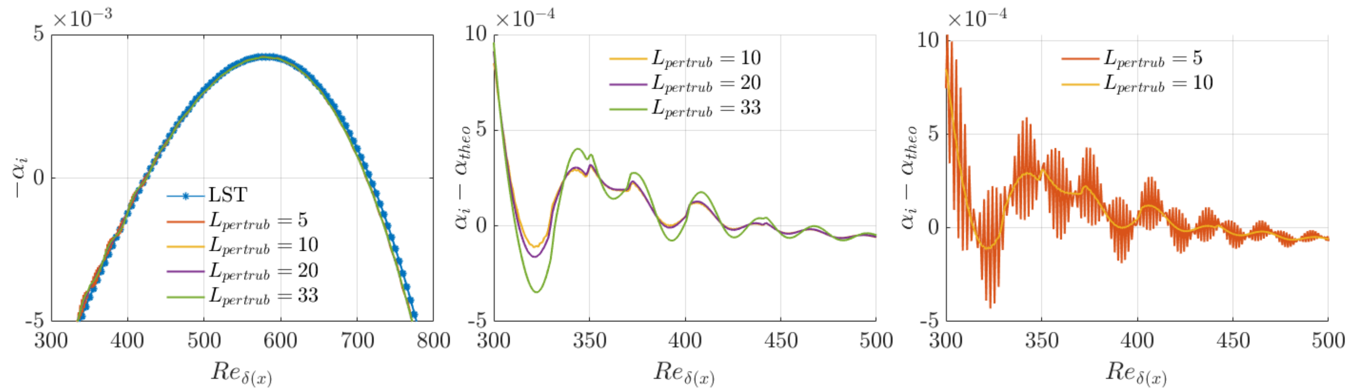


FIGURE 3.8: From left to right: Growth rate; Growth rate deviation from theory for $L_{perturb} = 10, 20$ and 33; Growth rate difference for $L_{perturb} = 5, 10$

Influence of the forcing position

Apart from forcing function and forcing length, another relevant parameter is the forcing position. Fixing other factors (using original forcing function and $L_{perturb} = 20$), 3 forcing positions are used to test the possible effects of the forcing position. Growth rates and differences between DNS results and LST analysis are shown in figure3.9. It is evident that the $Re_{perturb} = 150$ result gives the largest oscillation, and oscillation amplitude difference between $Re_{perturb} = 250$ and $Re_{perturb} = 300$ is not very significant. This may be because that there is more space for TS waves to decay, but the other mode do not decay or decay more slowly than TS waves.

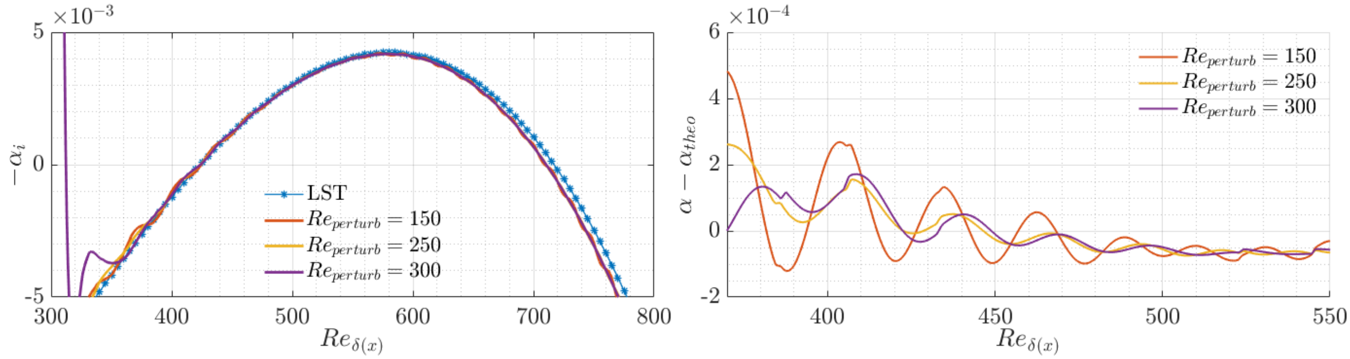


FIGURE 3.9: Growth rate (left) and growth rate deviation from theory (right) for different locations of the forcing, $Re_{perturb}$.

Conclusion of the upstream oscillations discussion

Three principal factors of the forcing: forcing function, forcing length and forcing position, are studied to discuss their impacts on the upstream oscillations. This also helps to discover the nature of these oscillations and eventually find a possible way to reduce it. Comparison between results using difference forcing functions shows that the original forcing function is good enough to give smooth curve of the growth rate and larger difference between forcing function and T-S waves amplifies the oscillation, which partly confirms our assumption that the oscillations come from another mode. It is also found that smaller forcing length and relatively downstream forcing position could possibly reduce its amplitude. However, the true nature of this oscillation is still unclear and we do not exclude the possibility that it is numerical instead of physical originated.

3.2 Validation of the subharmonic secondary instability

In this section, we would like to start the first studies and discussions about 3D instabilities, also to validate our post-processing methods and scripts for simulation data in 3D configuration, by recovering the results of Herbert (1988), Kachanov and Levchenko (1984) and Sayadi et al. (2013) for H-type secondary instability.

The parameters defining this simulation are specified with the case name **3D IG H-type** in table A.1. These values take reference of Sayadi et al. (2013), and are adjusted to the DNS program used in this study. Both Kachanov and Levchenko (1984) and Sayadi et al. (2013) use the amplitude of u' at fixed wall normal position $y^*/\delta^* = 0.26$ as instability measurement. For this reason, we use the criterion (d) in table 2.3 as reference rather than the criterion (c). Therefore, here the fixed y position in criterion (d) is precisely 0.26 and the value presented by criterion (c) is actually $\sqrt{\int_0^{L_y} |u'|^2 dy}$ scaled to match the value of (d) at forcing position.

Figure 3.10 shows the growth curves of the three criteria. For primary instabilities, differences between different criteria is small. The secondary instability curves are more different, but the evolution of these two curves is similar. Curves of all three criteria follow the analytical prediction from Herbert (1988). We

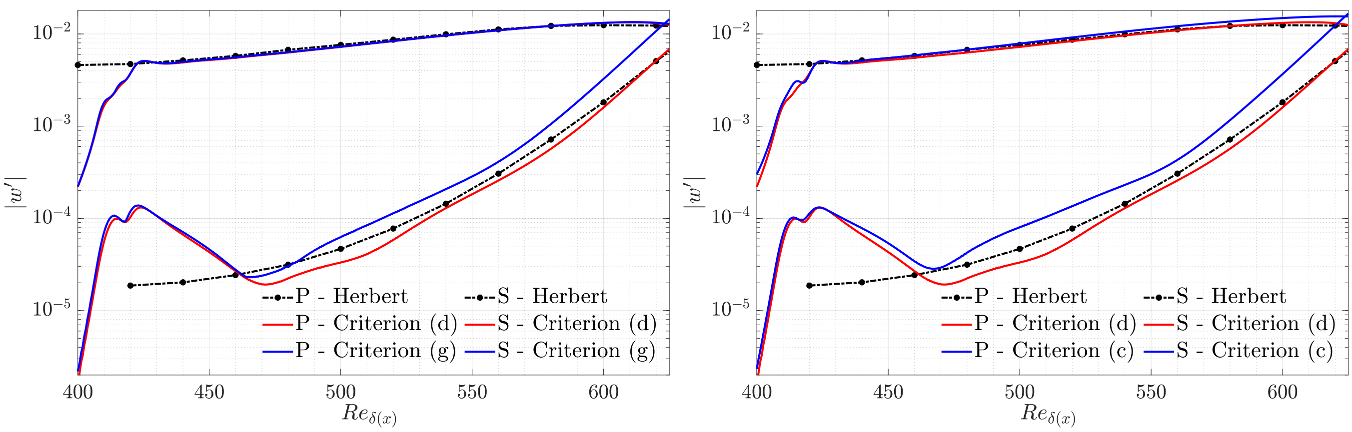


FIGURE 3.10: Amplitude of primary (noted as 'P') and secondary (noted as 'S') instabilities for H-type, compared with theoretical results from Herbert (1988). Left: comparison between results of criterion (d) and (g). Right: comparison between results of criterion (d) and (c).

note that the growth rates are not calculated here, because the amplitude curves, especially for secondary instability, are already slightly sinuous, which will be more prominent by growth rate calculation, making the results hard to analyze.

Apart from the growth tendency, fluctuation profile also contributes to the validation in 3D configuration. In figure 3.11, $|u'|$ profile found by the Fourier component of secondary mode using DNS data is compared with the theoretical results from Herbert (1988) and measurements from Kachanov and Levchenko (1984). All profiles are derived at position $Re_{\delta(x)} = 608$ and normalized by its maximum value. We again observed good agreement between different results. All results validate the numerical set-up and the post-processing methods to identify secondary instabilities.

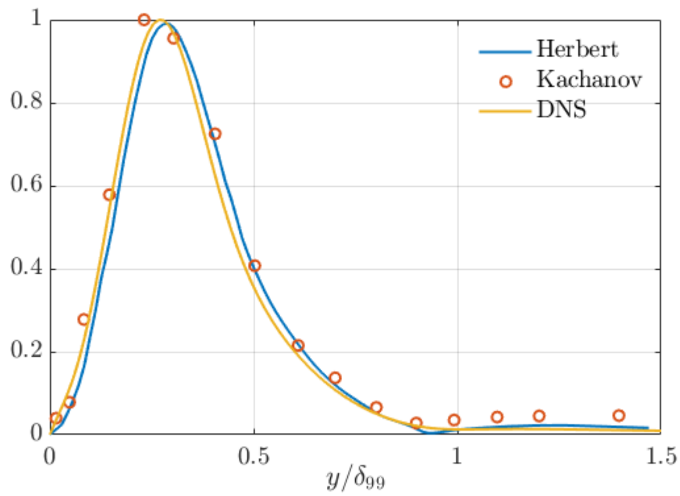


FIGURE 3.11: Normalized streamwise velocity fluctuation of subharmonic mode at $Re_{\delta(x)} = 608$; compared with theoretical results of Herbert (1988) and experimental results of Kachanov and Levchenko (1984)

Chapter 4

Real gas simulations with VdW EoS

After having performed ideal gas simulations, we introduce Van der Waals (VdW) equation of state (EoS) in the DNS simulations to model real-gas effects. Focusing on flow behaviors near the critical point, we restrict the studied cases with fixed thermodynamic parameters in table 4.1 taking reference from the work of Ren et al. (2019b).

Parameter	Definition	Value
Pr	Prandtl number	1
p^*_∞/p^*_c	Reduced freestream pressure (by critical pressure)	1.08
T^*_∞/T^*_c	Reduced freestream temperature (by critical temperature)	0.92
T^*_w/T^*_∞	Dimensionless wall temperature (by freestream temperature)	1.18
c_v/R	specific heat/gas constant proportion	5

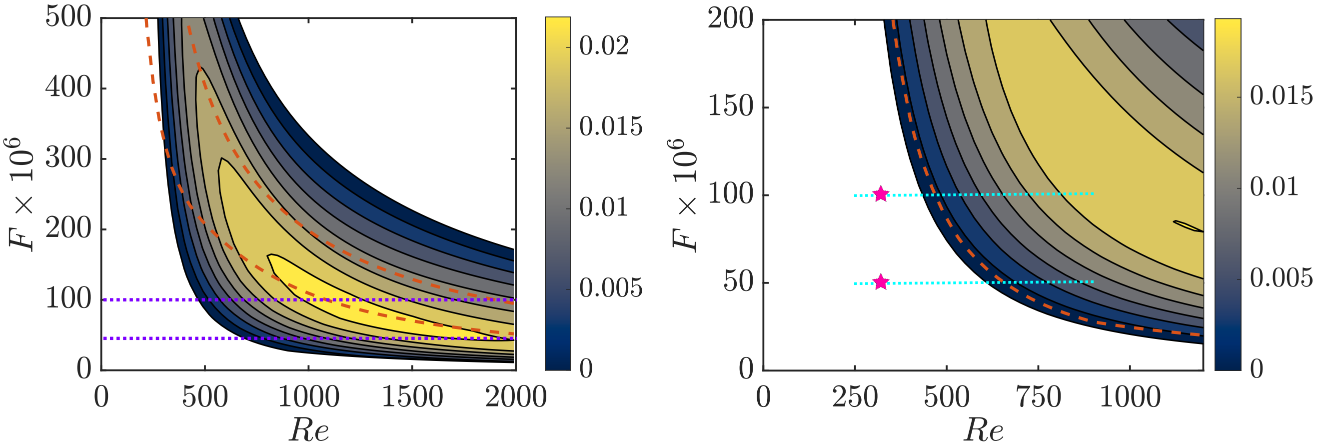
TABLE 4.1: Thermal dynamic properties used in VdW simulations

For a plat plate boundary layer, pressure is nearly constant over the whole domain, related to p^*_∞ imposed by simulation input. For the fluid under consideration, when $p^*/p^*_c = 1.08$, the pseudo-critical temperature value is $T^*_{pc}/T^*_c = 1.011$ after non-dimensionlized by T^*_∞, $T_{pc} = T^*_{pc}/T^*_\infty = 1.1$. Thus, when the dimensionless wall temperature $T_w > 1.1$, the boundary layer is in transcritical condition mentioned in section1.3. The wall temperature is fixed to $T_w = 1.18$, so the boundary layer is transcritical state and Mode II can appear.

4.1 Study of Mode II

In this part small forcing amplitude is used to trigger Mode II in linear regime. Figure 4.1 shows the stability diagram derived from LST with $T_w = 1.18$. figure 4.1a is found in the limit of zero Eckert number, the dashed red line show the contour of Mode I. The dimensionless frequency $F = 100 \times 10^{-6}$ is chosen again, and the subharmonic frequency $F = 50 \times 10^{-6}$ is also tested. The dashed purple line for $F = 100 \times 10^{-6}$ shows that only Mode II exists when Re_δ is roughly smaller than 1000. Figure 4.1b is the stability diagram calculated with $Ec = 0.05$. The dashed red line shows the contour of $Ec = 0$. This demonstrates the small differences, thus DNS results found by $Ec = 0.05$ can still be of reference for incompressible flow behavior. The interval of $Re_{\delta(x)}$ and the forcing position chosen for DNS are also marked in the figures. This choice is made such that the growth is only due to Mode II.

The exact value of Re_δ range, forcing position and other DNS parameters are shown in tableA.1 with the case name **2D VdW REF**. Compared to ideal gas simulations, DNS settings change because of the characteristics of transcritical boundary layer and Mode II. Because thermodynamic and transport properties have strong variation around critical point, a larger stretching coefficient is used for ensuring a better refinement near the wall. As the growth rate of Mode II is one order of magnitude larger than TS waves,



(A) Stability diagram with Eckert number set to zero. (- - -): neutral curve of Mode I, (⋯⋯): frequency marker

(B) Zoomed stability diagram with $Ec = 0.05$. (- - -): contour of $Ec = 0$, (⋯⋯): span of the computational domain, ($\star$): forcing position.

FIGURE 4.1: $F - Re_\delta$ stability diagram in the case of $T_w = 1.18$

the domain height doubles and sponge length increased in order to reduce nonphysical oscillations caused by reflection. Additionally, the outlet sponge length is increased significantly because the computational domain ends at a position where Mode II is still unstable with a large growth rate.

For $F = 100 \times 10^{-6}$, A_1 defined in the forcing equation (2.29) is chosen to be 5×10^{-10}. The curve of maximum value criterion (g) approaches most to the LST result, but the difference between the 3 criteria and LST results is small, which validates the chosen computational parameters. Apart from criteria, we can see that the amplitude of this mode becomes 1000 times greater crossing the computational domain, but still stays with the linear prediction when $A_1 = 5 \times 10^{-10}$.

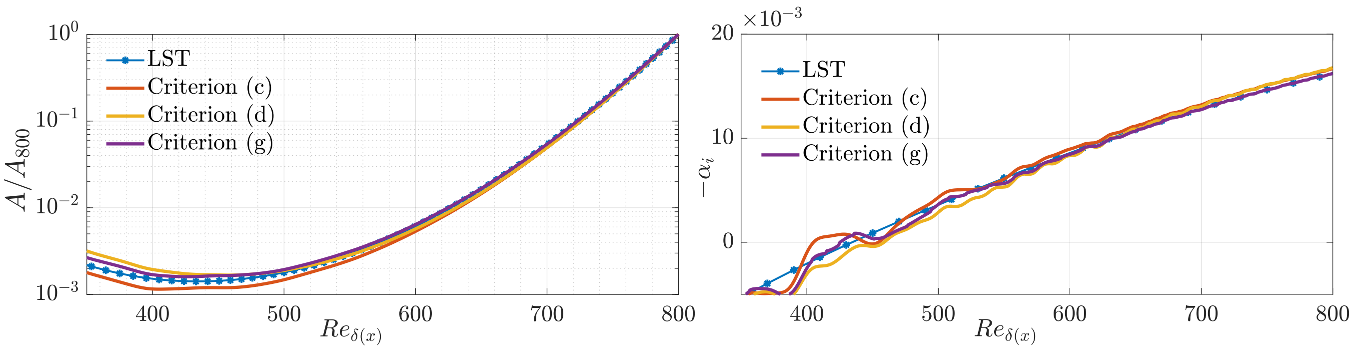


FIGURE 4.2: Amplitude (left) and related growth rate (right) of $F = 100 \times 10^{-6}$ using criteria in table 2.3. Amplitude values are normalized at $Re_\delta = 800$.

After the first case, another simulation with the subharmonic frequency $F = 50 \times 10^{-6}$ is conducted, in purpose of studying the instability behavior of Mode II at this frequency, which will be compared with the subharmonic secondary instability growth of $F = 100 \times 10^{-6}$. In this case, different A_1 are tested. When A_1 is as small as 5×10^{-8} there are more oscillations in the amplitude curves than that are presented in figure 4.3, so as to the growth rate curves. It is probably due to the fact that this mode is stable in most part of this Re_δ range, thus the amplitude becomes so small that numerical fluctuations influence the results. When A_1 increases from 5×10^{-6} to 5×10^{-5}, the curves stay unchanged. This observation limits our choice of A_2 for studying secondary instabilities of primary frequency $F_1 = 100 \times 10^{-6}$. Results independent of forcing amplitude are presented in figure 4.3. The amplitude curves of all three criteria follow LST prediction, but compared to the previous case with $F = 100 \times 10^{-6}$, the differences become larger. The maximum value criterion (g) and the integration criterion (c) fit better to LST prediction than the fixed y criterion (d). For the growth rate, oscillations are observed for all three criteria but the tendency still corresponds to LST

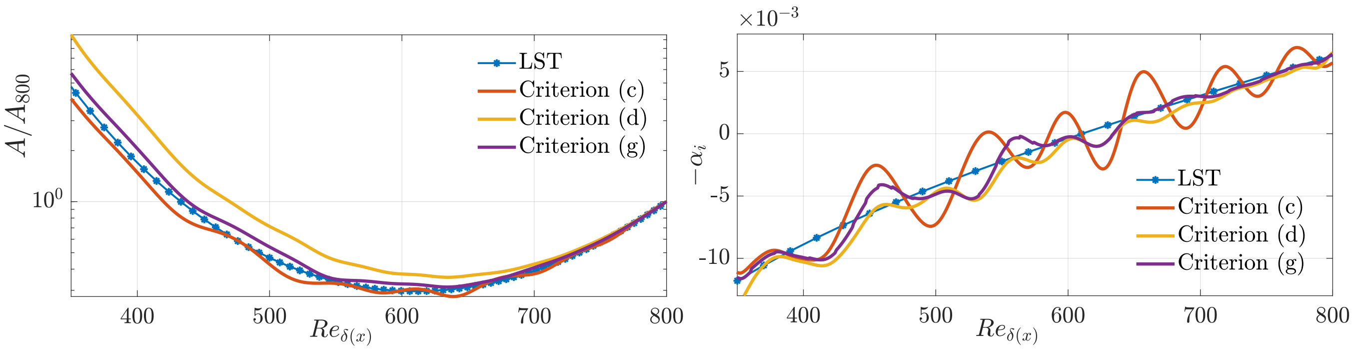


FIGURE 4.3: Amplitude (left) and related growth rate (right) of $F = 50 \times 10^{-6}$ using criteria in table 2.3

prediction. Because of these oscillations, only amplitude curves will be used to evaluate the growth of each mode in the following parts.

On the other hand, in 3D simulations, fix y criterion (d) is preferred because it can save storage. To find the optimum wall normal position, fixed position criterion with different positions are tested for both frequencies and the results are shown in figure 4.4. The amplitudes are normalized at the forcing position. The plots show that fixed y criterion is less dependent on the chosen position in $F = 100 \times 10^{-6}$ case than the $F = 50 \times 10^{-6}$ case. For both cases, y position around 0.30 can give the most consistent results with LST.

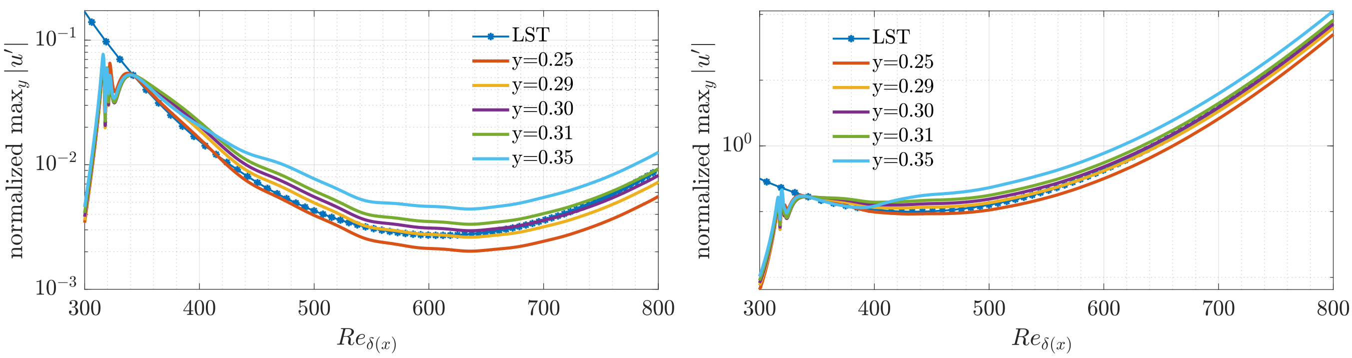
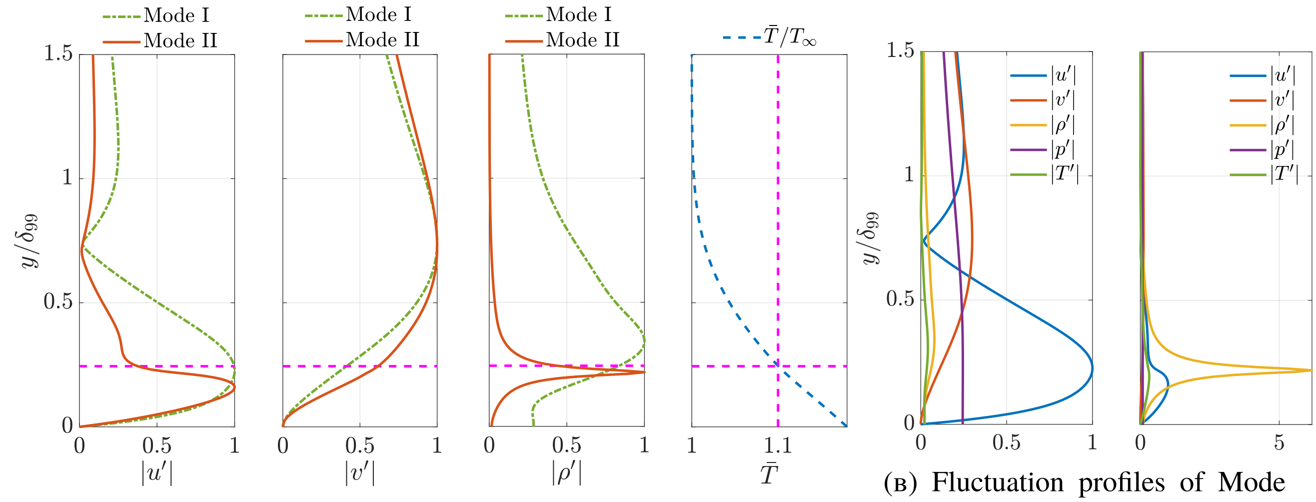


FIGURE 4.4: Amplitude using criterion (d) with different y position for $F = 50 \times 10^{-6}$ (left) and $F = 100 \times 10^{-6}$ (right)

Apart from the amplitudes and growth rate, we also would like to know how different the fluctuation profiles of Mode I and Mode II are. Figure 4.5a shows the streamwise velocity fluctuation $|u'|$, normal velocity fluctuation $|v'|$ and density fluctuation $|\rho'|$ profiles derived by DNS of the two modes.The mean profile of the temperature is also added which helps to mark the position where temperature reaches to the pseudo-critical temperature. Mode I profiles are found by the simulation mentioned in section 3.1. All curves are plotted at $Re_\delta = 500$ with $F = 100 \times 10^{-6}$, but considering self-similarity and insignificant parallel effects, these results can be generalized to other positions and frequencies. There is some similarity in velocity fluctuation profiles between the 2 modes, the differences are closely related to the position where the temperature reaches to the pseudo-critical temperature. The density profile changes most significantly, it has a sharp maximum value near the position of T_{pc}. These changes are expected because of the strong variation of thermodynamic and transport properties around pseudo-critical point while compressible and thermal effects are negligible in the ideal gas simulation. Figure 4.5b shows the fluctuation profiles of Mode I (left) and Mode II (right). Profiles of $|u'|$, $|v'|$, $|\rho'|$, $|p'|$ and $|T'|$ are normalized by the maximum value of $|u'|$. Compared to Mode I, the density fluctuation of Mode II has greater amplitude than other fluctuations, this is has been found by Ren et al. (2019b) and helps us to confirm again the validation of DNS results.



(A) From left to right: fluctuation profiles at $Re_\delta = 500$ of u, v, ρ of Mode I and Mode II, normalized by its maximum value, and the mean profile of T. (- - -): position where $T = T_{pc}$.

(B) Fluctuation profiles of Mode I (left) and Mode II (right), normalized by the maximum value of streamwise velocity fluctuation.

FIGURE 4.5: Profiles of mean flow and perturbations

4.2 2D secondary instability of Mode II

4.2.1 Detection of a subharmonic instabliity

After recovering Mode II, we would like to find the subharmonic secondary instability in 2D configuration. Although 3D configuration is used to study secondary instabilities in the case of ideal gas, we start with 2D configuration with supercritical fluids because the secondary instability of Mode II is likely to grow in 2D and 2D simulations take less time to accomplish thus allow us to speed up the discovery by trial and error.

The primary frequency is chosen to be $F_1 = 100 \times 10^{-6}$ and the subharmonic forcing is added by defining $F_2 = 50 \times 10^{-6}$ and $A_2 = 5 \times 10^{-6}$ in equation (2.29). These values are fixed while several different values of A_1 are used. As the criterion (g) is shown to correspond best to LST prediction in the previous section, only this criterion is used in this part.

To start with, one simulation with particularly small value of A_1 ($A_1 = 5 \times 10^{-8}$) and another with relatively large A_1 ($A_1 = 10^{-3}$) value are conducted, the amplitude curves are respectively shown on the left and right of figure 4.6. For small A_1 case, the growth of both frequencies follows LST prediction, so the behavior of subharmonic frequency is totally due to Mode II at $F = 50 \times 10^{-6}$, there is no secondary instability growth. Secondary instabilities appear when A_1 is large enough. With large A_1 (shown on the right of figure 4.6), the $F = 50 \times 10^{-6}$ curve changes, indicating that it becomes more unstable in the latter part of the domain.

It is validated in the previous chapter that only 4 samples are needed to find the smooth curve that we would like to extract from DNS, and it still holds for the $A_1 = 5 \times 10^{-8}$ case, but more samples are needed in the $A_1 = 10^{-3}$ case. Figure 4.7 shows the primary and secondary instability curves found by using different number of samples in one period. When smaller numbers of samples are used, the end part of secondary instability curve becomes very messy. This demonstrates that even using 18 samples in one period is not enough, 36 samples in one period are needed to give the expected growth curve. In the following parts, more number of samples are used in simulations with large A_1 such that the growth curves do not depend on sampling parameters.

In secondary instability analysis, the growth rate of secondary instabilities is related to the amplitude of primary instability. To study this effect, different A_1 are used and results are plotted in figure 4.8 along with LST prediction of Mode II. On the left of figure 4.8, the amplitudes of primary instability with different A_1 are normalized to have the same value at forcing position to facilitate the observation of primary instability change. The corresponding subharmonic growth is plotted on the right of figure 4.8. The first observation is

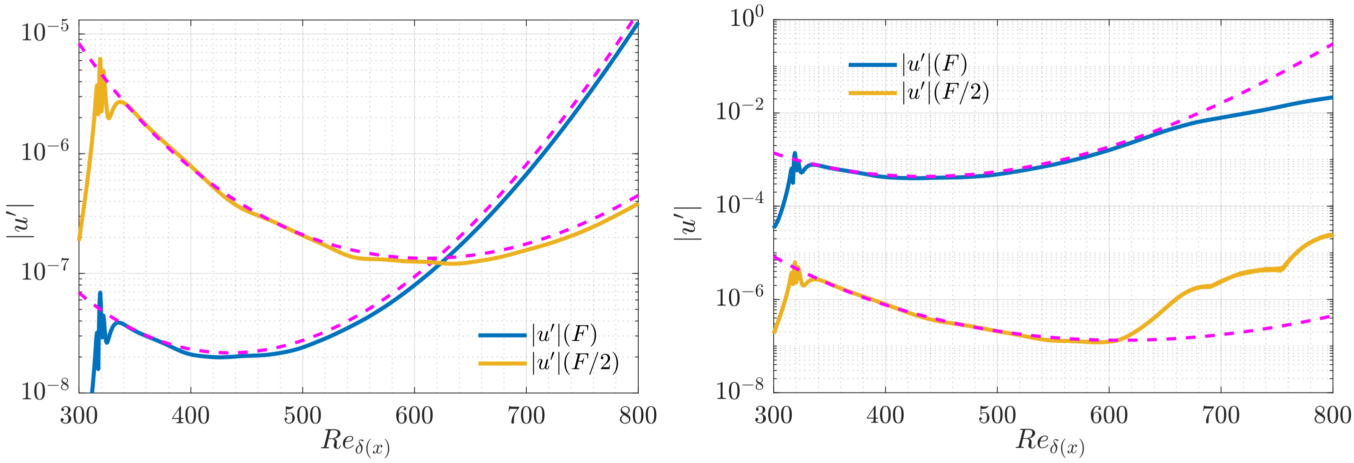


FIGURE 4.6: Amplitude evaluation of $F = 100 \times 10^{-6}$ and $F/2 = 50 \times 10^{-6}$ with $A_1 = 10^{-8}$ (left) and $A_1 = 10^{-3}$ (right). (- - -): LST prediction of Mode II.

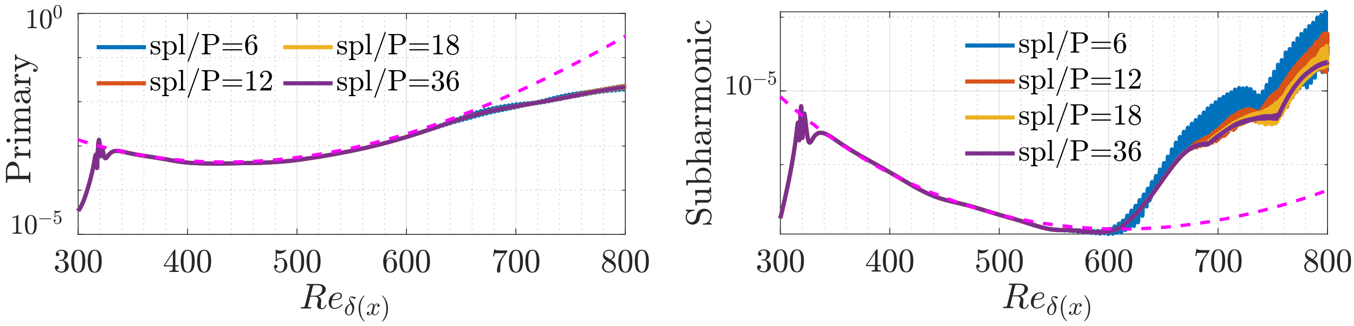


FIGURE 4.7: Primary (left) and subharmonic (right) instability growth of case $A_1 = 10^{-3}$ using different number of samples in one period over 2 period of the primary instability. (- - -): LST prediction of Mode II.

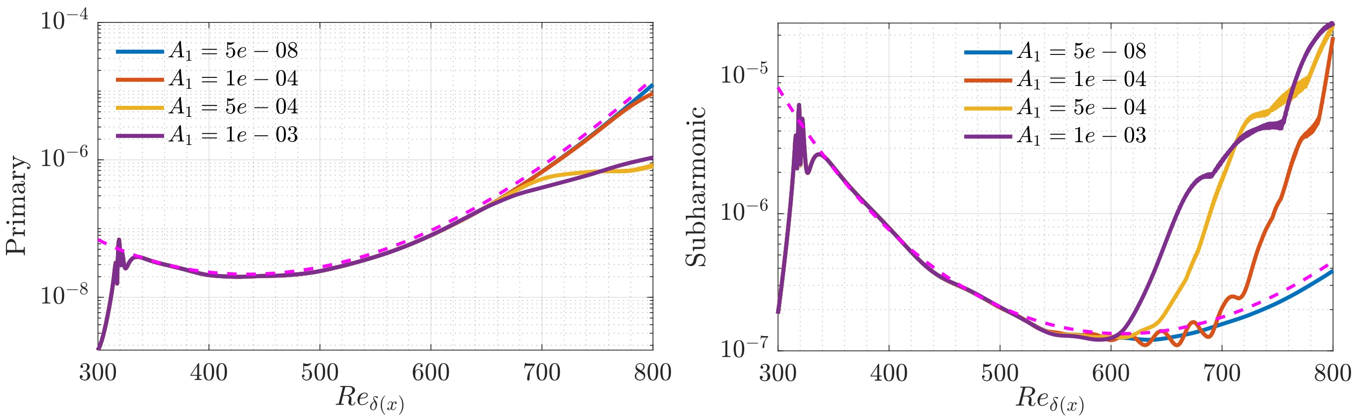


FIGURE 4.8: Left: normalized primary instability amplitude with different values of A_1. Right: the corresponding subharmonic instability growth. (- - -): LST prediction of Mode II.

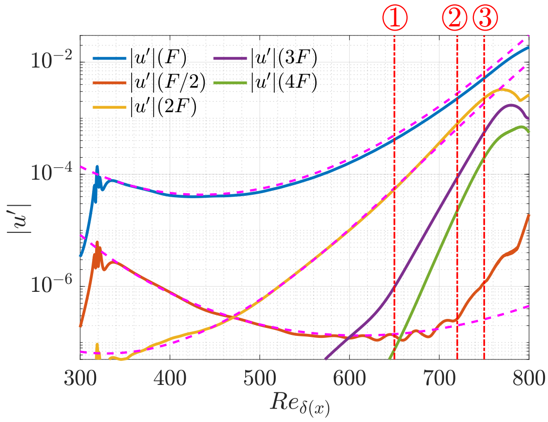
that the secondary instability starts to grow earlier and faster for lager A_1, which is consistent with secondary instability theory. Except the one with smallest A_1 value, secondary instability growth exists in all other three cases. For these three cases, larger values of A_1 ($A_1 = 5 \times 10^{-4}$ and 10^{-3}) give primary instability curves that change at the end of the domain. This is different from the H-type secondary instability in ideal gas, in which the primary instability satisfies linear assumption. The change of primary instability growth can be explained by non-linear behaviors and resonant interactions. We especially note the $A_1 = 10^{-4}$ case, its primary instability curve is not distorted and the secondary growth also exists. Another particularity of

$A_1 = 10^{-4}$ case is the oscillations at the position where secondary growth appear, this may be related to the transition from Mode II growth to subharmonic secondary growth, for larger A_1, the transition happens faster thus less observable.

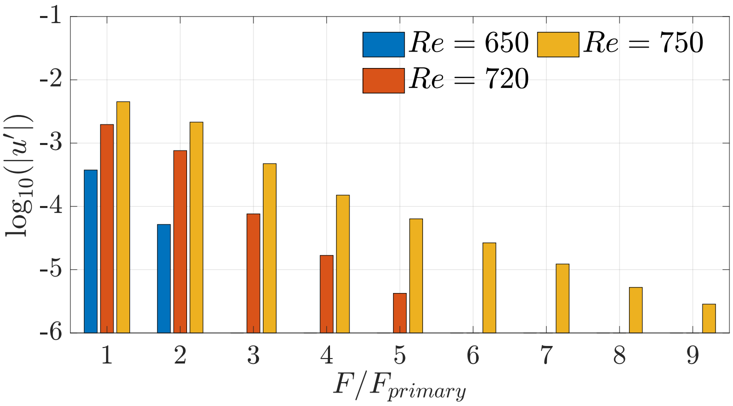
To investigate further into these simulations and try to prove the existence of non-linear effects, we study the instability behaviors from the following aspects: the response of higher harmonics, the fluctuation profiles and the mean profiles. To compare with ideal gas H-type instability, the 3D ideal gas simulation in section 3.2 is also included and is analyzed in the same manner.

4.2.2 Response of higher harmonics

After observing the change of the primary instability curve, we would like to analyze the non-linear effects by studying the behavior of higher harmonics. Growth curves and spectra at different positions are examined. Two VdW simulations ($A_1 = 10^{-3}$ and $A_1 = 10^{-4}$), in which secondary instability growth exists, are chosen to be studied. We recall that the primary curve in $A_1 = 10^{-3}$ case does not follow LST prediction at the end of the domain but the primary instability satisfies LST prediction for the $A_1 = 10^{-4}$ case. In the figure of growth curves, we limit the highest harmonics to $4F$. Spectra at several positions are potted to demonstrate the amplitudes of even higher harmonics. Figure 4.9 and figure 4.10 show respectively the results of case $A_1 = 10^{-3}$ and case $A_1 = 10^{-4}$. Figure 4.12 shows the results of the ideal gas simulation in section 3.2.



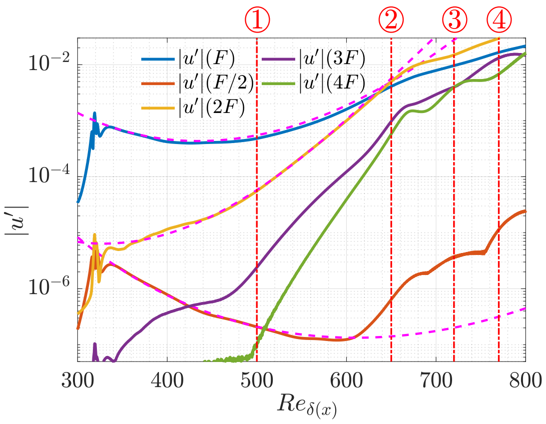
(A) Growth of different harmonics using criterion (g). (- - -): LST results, (-·-·-): position markers.



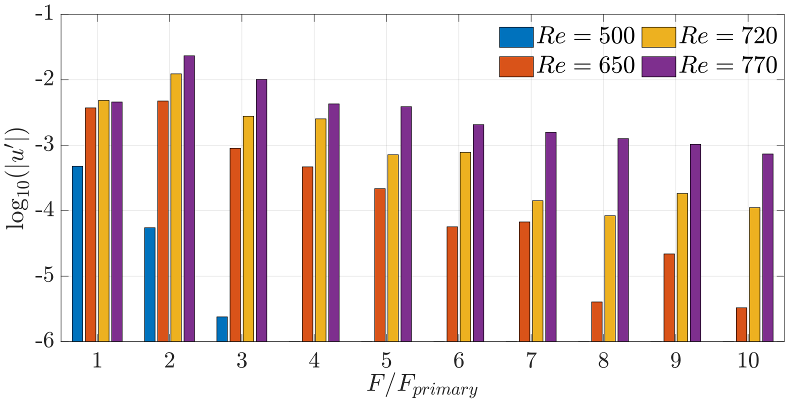
(B) Spectrum of different harmonics at $y = 0.3$ and the four streamwise positions marked on figure 4.9a

FIGURE 4.9: Spectrum of transcritical VdW simulation with $A_1 = 10^{-4}$ at three positions: ① Re=650, ② Re=720, ③ Re=750 with $y = 0.30$. The primary instability frequency $F = 100 \times 10^{-6}$.

Compare to the ideal gas simulation, higher harmonics in VdW simulations grow much faster and have larger amplitudes. This explains why more samples are needed, as sub-sampling leads to information lost so the results become inaccurate. For both VdW cases, the growth of subharmonic frequency increases but is significantly smaller than the higher harmonics, which is not the case for ideal gas simulation. While the primary instability still has the largest amplitude in $A_1 = 10^{-4}$ case, the amplitude of harmonic $2F$ exceeds that of primary instability near the end of the domain in $A_1 = 10^{-3}$ case (shown in figure 4.10), which may explain the growth change in primary instability. LST prediction of Mode II for $2F$ is also plotted in figure 4.9 and 4.10, and the DNS results well conforms LST. However, it is not the case for harmonics higher than $2F$. In figure 4.11, growth curves of $3F$ and $4F$ in $A_1 = 10^{-3}$ case are plotted with LST results, showing their significant differences, especially for the harmonic $4F$. As we would like to know what is origin of these differences from LST for higher harmonics and also for ensuring that they are numerical errors, several other simulations are conducted and the results are discussed in a separated part.



(A) Growth of different harmonics using criterion (g). (- - -): LST results, (-·-·): position markers.



(B) Spectrum of different harmonics at $y = 0.30$ and the four streamwise positions marked on figure 4.10a

FIGURE 4.10: Growth curves and spectrum of transcritical VdW simulation with $A_1 = 10^{-3}$ at three positions: ① Re=500, ② Re=650, ③ Re=720, ④ Re=770 with $y = 0.30$. The primary instability frequency $F = 100 \times 10^{-6}$.

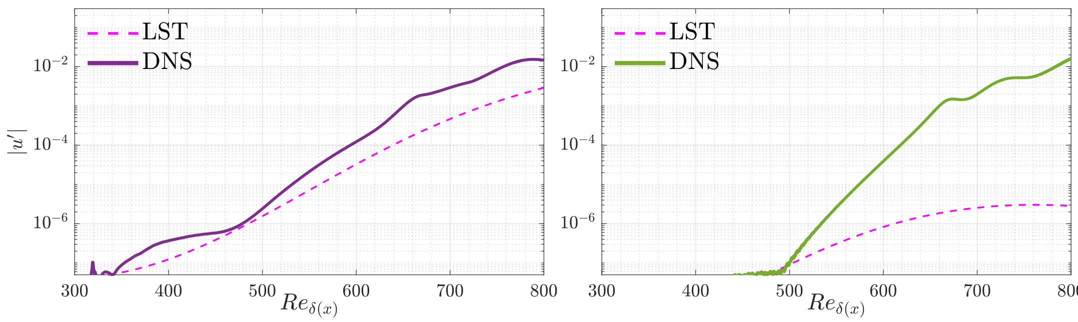
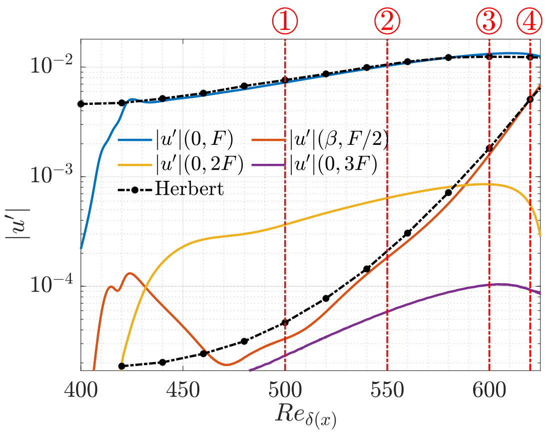
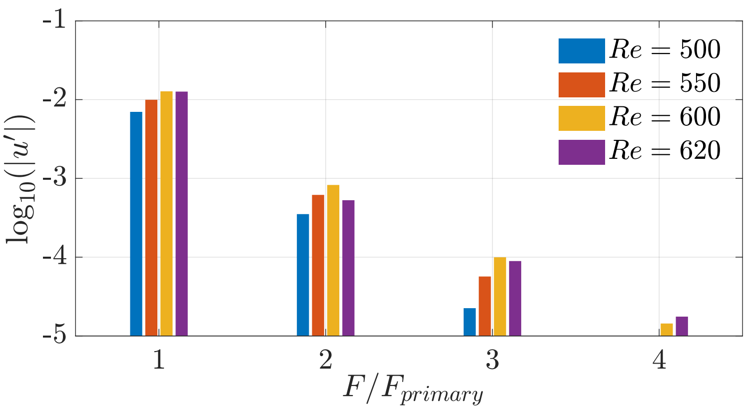


FIGURE 4.11: Growth of harmonics $3F$ (left) and $4F$ (right) using criterion (g), with $F = 100 \times 10^{-6}$ and $A_1 = 10^{-3}$, compared with LST analysis.



(A) Amplitude of different harmonics using criterion (g). (-·-·): position markers.



(B) Spectrum of different harmonics at $y = 0.26$ and the four streamwise positions marked on figure 4.12a

FIGURE 4.12: Growth and spectrum of ideal gas H-type validation case at four positions: ① Re=500, ② Re=550, ③ Re=600, ④ Re=620 and $y = 0.26$. The primary instability frequency $F = 124 \times 10^{-6}$.

The spectra in figure 4.9b and 4.10b further confirm the large amplitudes of higher harmonics in VdW simulations. For both VdW simulations, the amplitudes grow larger as the flow develops downstream. For

$A_1 = 10^{-3}$ case, the amplitudes of higher harmonics are much greater than the $A_1 = 10^{-4}$ case, especially at the last chosen position, where the amplitude of harmonic $10F$ is only one magnitude smaller than the primary mode. Along with the mismatching growth between DNS results and LST prediction with $3F$ and $4F$, we can conclude that non-linear effects are not negligible. Compare to the higher harmonics, the contribution of subharmonic to the total disturbance is tiny, so in this case the transition to turbulence may be not due to the secondary instability growth, but non-linear resonance interaction of the instabilities.

4.2.3 Mean and fluctuation profiles

Apart from growth curves and spectra, mean and fluctuation profiles are also good indicators of instability behaviors and disturbance levels. Figure 4.13 and 4.14 shows the mean profiles of u, ρ and T for $A_1 = 10^{-4}$ case and $A_1 = 10^{-3}$ case. In figure 4.13, the mean profiles of all three variables keep the same form at the three positions marked in figure 4.9. In $A_1 = 10^{-3}$ case (figure 4.14), the mean profiles deforms at different positions because of the non-linear effects, especially for the density. For more upstream positions where non-linearities are not yet significant, the density profile has a larger gradient near the position where $T = T_{pc}$, but it becomes smoother at downstream positions as the fluctuation amplitude grows and non-linearity enters into the problem.

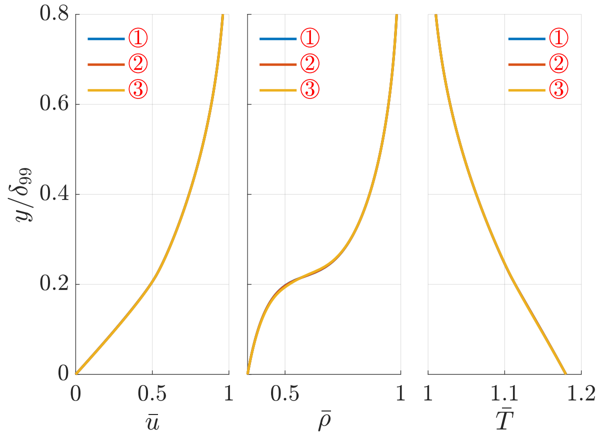


FIGURE 4.13: Mean profile of u, ρ, T for VdW simulation with $A_1 = 10^{-4}$

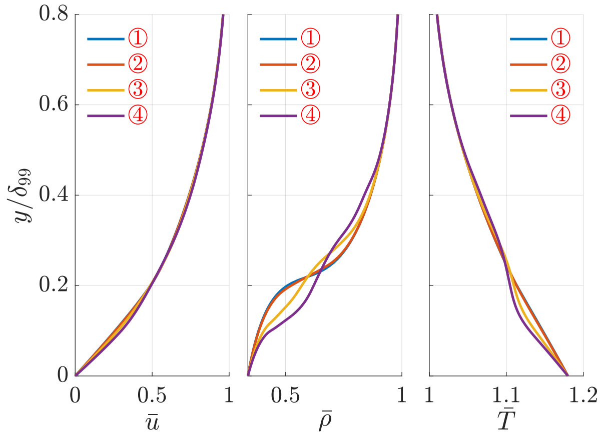


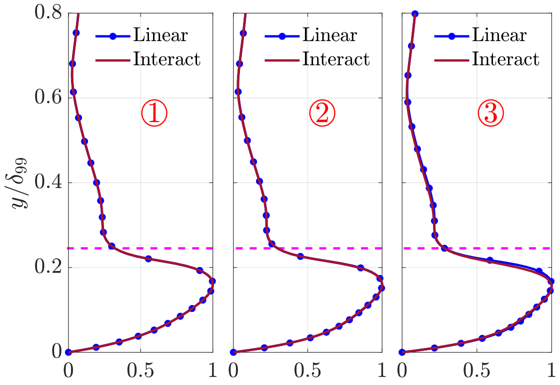
FIGURE 4.14: Mean profile of u, ρ, T for VdW simulation with $A_1 = 10^{-3}$

Next, primary and subharmonic Fourier components are plotted for variable u, v and ρ. For simplicity reason, only profiles of $|u'|$ are presented in this part and the other plots can be found in appendix C. For observing more clearly the changes bought by secondary instability growth or non-linear effects, profiles in the $A = 5 \times 10^{-8}$ case are used as reference of linear results at each position. Figure 4.15 shows the profiles for $A_1 = 10^{-4}$ case at the three chosen positions marked on figure 4.9. We recall that the primary instability is not distorted in this case. The good match to the linear reference profiles further confirms that the primary instability still follows the linear prediction. For the subharmonic instability, the profile stays the same as the linear profile at $Re = 650$, which is the position ①, and it deforms at the other two positions. There is some similarity between profiles at position ② and ③, and this may indicate what the secondary $|u'|$ profile look like when it is calculated by secondary instability analysis which is not conducted yet. More importantly, it can be an indicator of secondary instability for higher A_1 simulations in which non-linear effects are also present.

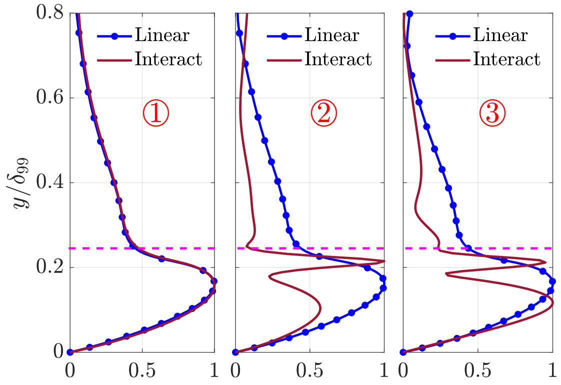
Figure 4.16 shows the profiles for $A_1 = 10^{-3}$ case at the four chosen positions marked on figure 4.10. In this case, the primary instability curve follow LST prediction at position ① and ② but not for the others. Coherently, the primary instability profiles well coincide with linear profile reference at these two positions. As the secondary instability is not growing at position ①, no deformation can be observed in the subharmonic profile. At position ②, where the primary instability growth is still linear and secondary instability appears, the subharmonic profile deforms and its shape is similar to the ones shown in the figure 4.15b.

For position ③ and ④, both primary and subharmonic profiles deform and there is no pattern that can be summarized. This is probably because of the disorderliness added by the non-linear effects. Therefore, the last two positions are not suitable to study secondary instability growth but non-linear interactions between different modes.

Although the previous discussions are based only on $|u'|$ profiles, we note that other fluctuation profiles also support the previous analysis.

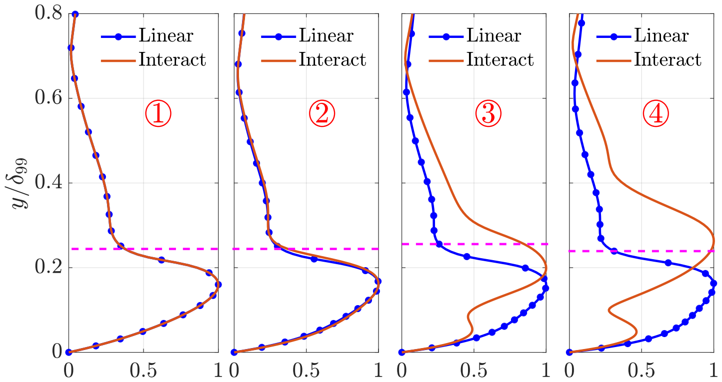


(A) Primary $|u'|$ profiles

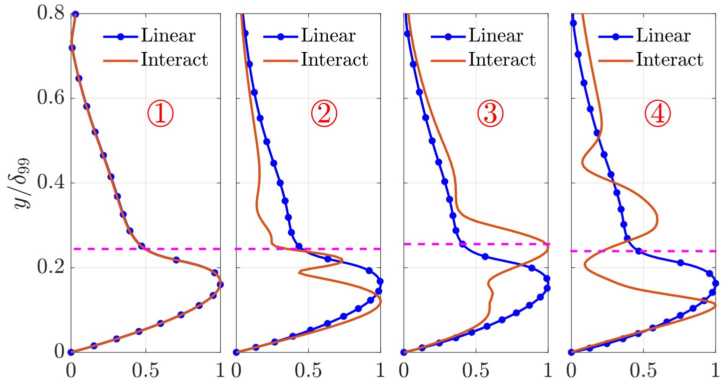


(B) Subharmonic $|u'|$ profiles

FIGURE 4.15: $|u'|$ profiles of VdW simulation with $A_1 = 10^{-4}$ at the four positions marked in 4.9a.. Values are normalized by the maximum value of each profile. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.



(A) Primary $|u'|$ profiles



(B) Subharmonic $|u'|$ profiles

FIGURE 4.16: $|u'|$ profiles of VdW simulation with $A_1 = 10^{-3}$ at the three positions marked in 4.10a. Values are normalized by the maximum value of each profile. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.

4.2.4 Interaction of higher harmonics

Interaction of higher harmonics are studied separately in this part because of the observation of the peculiar behaviors of higher harmonics in the previous part. To achieve this, same computational settings are used as the precious part and we vary the forcing parameters F_1, F_2, A_1 and A_2.

The first case is shown in figure 4.17, in which $F_1 = 200 \times 10^{-6}$, $A_1 = 10^{-3}$, $F_2 = 100 \times 10^{-6}$, $A_2 = 5 \times 10^{-6}$. This case is initially intended to study the secondary instability growth of $F = 100 \times 10^{-6}$ with primary instability frequency $2F = 200 \times 10^{-6}$. However, large non-linear growth of higher harmonics is observed and mode F is stabilized. The mode $2F$ itself also deviates from LST prediction when its amplitude is about 10^{-2}. This partially explains why in figure 4.10, when the amplitude of $2F$ exceeds F with amplitude value approximately equal to 10^{-2}, both modes F and $2F$ grow more slowly. Another remark is about LST prediction to mode $2F$, according to which the amplitude of mode $2F$ grows more than 10^5 times larger crossing the domain. This is due to the characteristic of Mode II, and this rapid growth can quickly trigger non-linear effects.

The next cases with higher harmonics involves $3F$ and $4F$. These two frequencies are chosen because of their non-linear growth observed in figure 4.11. Three simulations are accomplished. The first one used $F_1 = 300 \times 10^{-6}$, $F_2 = 400 \times 10^{-6}$, $A_1 = A_2 = 5 \times 10^{-7}$, the results are shown in figure 4.18. The other two involves only one of those frequencies with $A_1 = 5 \times 10^{-7}$, $A_2 = 0$. The growth curve of $3F$ and $4F$ are respectively plotted on the left and right side of figure 4.19.

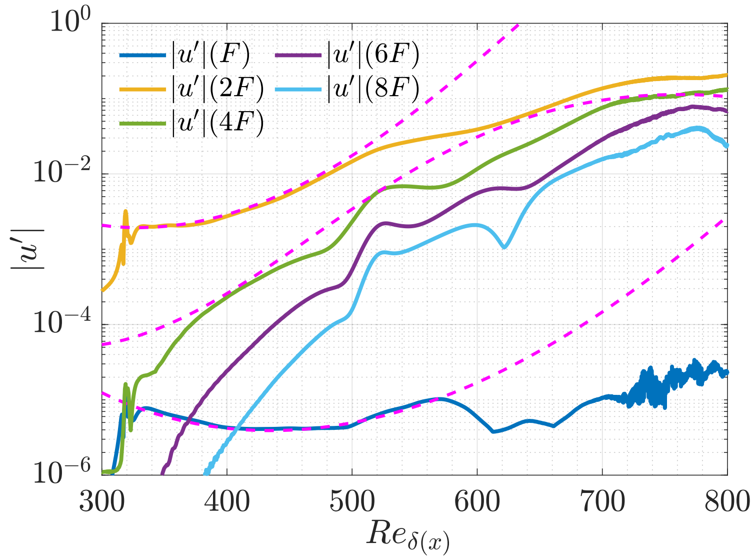


FIGURE 4.17: Interaction between $F_1 = F = 100 \times 10^{-6}$ and $F_2 = 2F = 200 \times 10^{-6}$ with $A_1 = 10^{-3}$ and $A_2 = 5 \times 10^{-6}$. (- - -): LST prediction.

For the two mono-harmonic simulations, only the growth of $3F$ is broken in the downstream region. Like mode $2F$, the amplitude of mode $3F$ also grows 10^5 times larger crossing the domain by LST prediction, and the amplitude of mode $3F$ at the position where DNS result no longer matches to the one of LST is also around 10^{-2}. For the mode $4F$, the growth follows better LST prediction with a slight mismatching which could be due to the non-parallel effects at higher frequency. This may be because the mode $4F$ has smaller growth rate and its amplitude has not yet reach to 10^{-2}, which seems to be a critical value for the appearance of significant non-linear effects. In the case involving the two frequencies, the growth curves of $3F$ and $4F$ are both deformed at the end of the domain, which may be only due to the non-linear effects of mode $3F$.

Although modes with other frequencies are not triggered in the mono-harmonic simulations and both modes behave linearly in the front part of the domain, modes F and $2F$ appear with rapid non-linear growth in figure 4.18. This suggests that F and $2F$ are triggered by the non-linearity of the forcing and their growth may be due to resonance interaction. At the position where rapid growth of F and $2F$ begin to happen, the

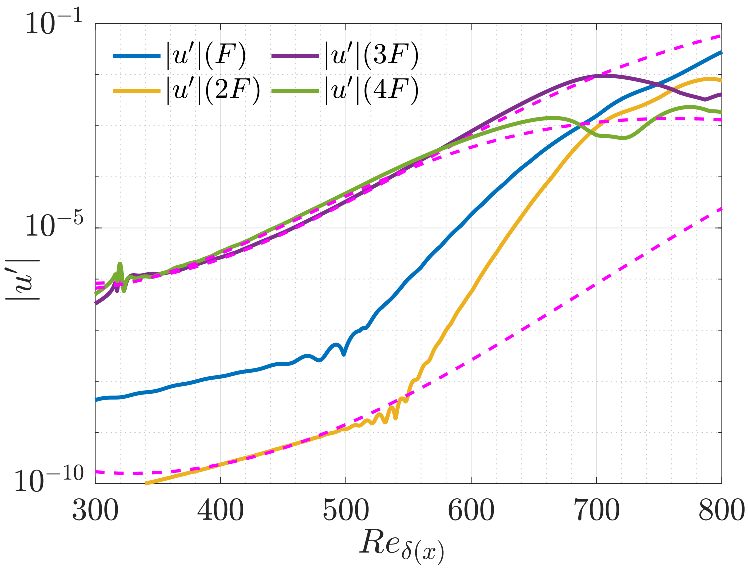


FIGURE 4.18: Interaction between $F_1 = 3F = 300 \times 10^{-6}$ and $F_2 = 4F = 400 \times 10^{-6}$, with $A_1 = A_2 = 5 \times 10^{-7}$. (- - -): LST prediction.

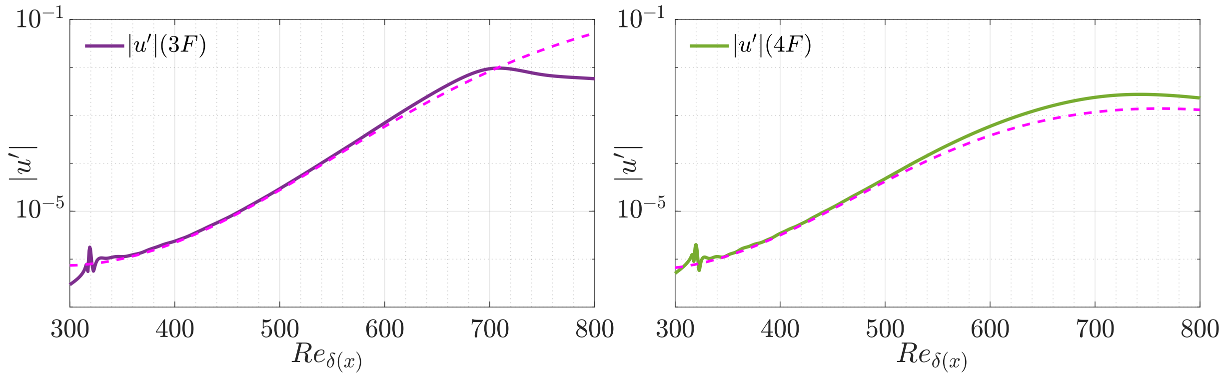


FIGURE 4.19: Mono harmonic growth of $3F$ (left) and $4F$ (right), with $F = 100 \times 10^{-6}$, $A = 5 \times 10^{-7}$. Growth of other harmonics is not observed thus omitted in the figure. (- - -): LST prediction.

amplitude of mode $3F$ and $4F$ have not reached to 10^{-4} yet, therefore the mechanism cause the growth of F and $2F$ does not require very large amplitude, and this can be one direction of exploration for further study.

4.3 3D secondary instability of Mode II

After finding secondary subharmonic growth in 2D configuration, we would like to continue the study to 3D and investigate the effects brought by adding the spanwise wave number. We chose to fix the forcing parameters as $F_1 = 100 \times 10^{-6}$, $A_1 = 10^{-3}$, $F_2 = 50 \times 10^{-6}$ and $A_2 = 5 \times 10^{-6}$, which are the same as the case with growth curves shown on figure 4.10a, because the secondary instability growth is larger in this simulation which is better for observing changes. Because 3D simulations require more computational power and take more time to complete, a reduced mesh is used first to find a spanwise wave length that has perceptible influence on the secondary instability growth. More precisely, we begin with a very large spanwise length and gradually reduce it until change can be observed in the secondary instability curve. As it mentioned before, criterion (d) in table 2.3 is used for the sake of saving storage, and the value of fixed y position is 0.3, validated in section 4.1.

Because the mesh is reduced mainly in streamwise and wall normal direction, a simulation with the corresponding reduced 2D mesh is carried out to demonstrate how much precision is lost. The exact values of parameters defining the reduced mesh are shown in table A.1 as 2D VdW RDU. Figure 4.20 compares the growth curve found by the original fine mesh and the reduced coarse mesh. The two curves of primary

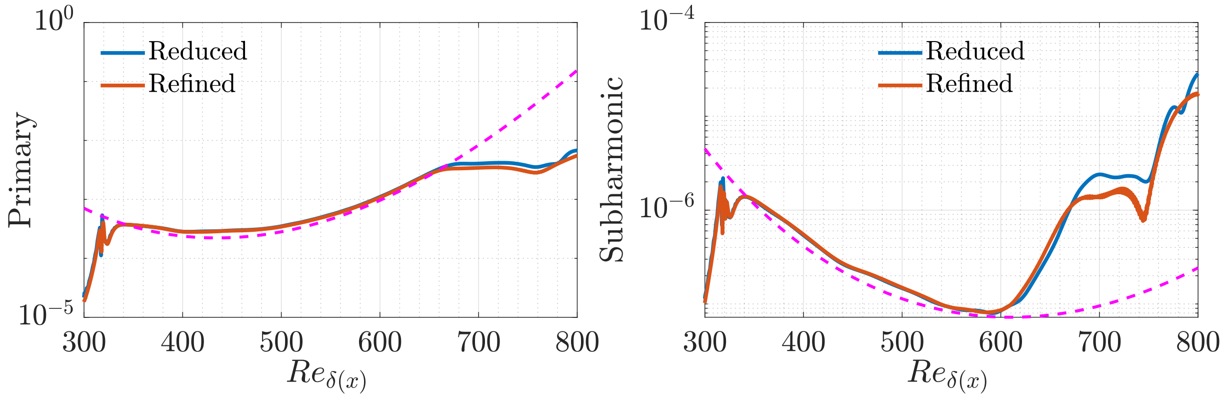


FIGURE 4.20: Primary (left) and subharmonic (right) instability growth using the original and reduced mesh and criterion (d). (- - -): LST prediction.

instability have little difference, but for the subharmonic curves, the difference is not negligible at positions where Re_δ is greater than 670. These are the positions where non-linear behaviors appear. Because non-linear effects can trigger flow motions that happen on smaller length scales and need finer mesh to capture. Therefore, we should note that the simulations using reduced mesh are not accurate at the end part of the domain.

The values of parameters used for the first 3D DNS are shown in table A.1 with name 3D VdW RDU. Like the 3D ideal gas case, the wave length of forcing is set as same as the spanwise length, starting by $\lambda_y = 200$ and gradually reduced to 50. Figure 4.21 shows the growth curves with three different wave length. For the primary instability curve, no change is observed. For the secondary curves, differences come mostly from the end part of the domain which is already mentioned to be inaccurate for the coarse mesh, but we still can draw insight of stronger non-linear effects for smaller wave length. When λ_y is reduced further to 50, the program crashes. This is likely to be caused by the non-linear effects which require finer mesh to solve smaller scales and to take dissipation effects into account.

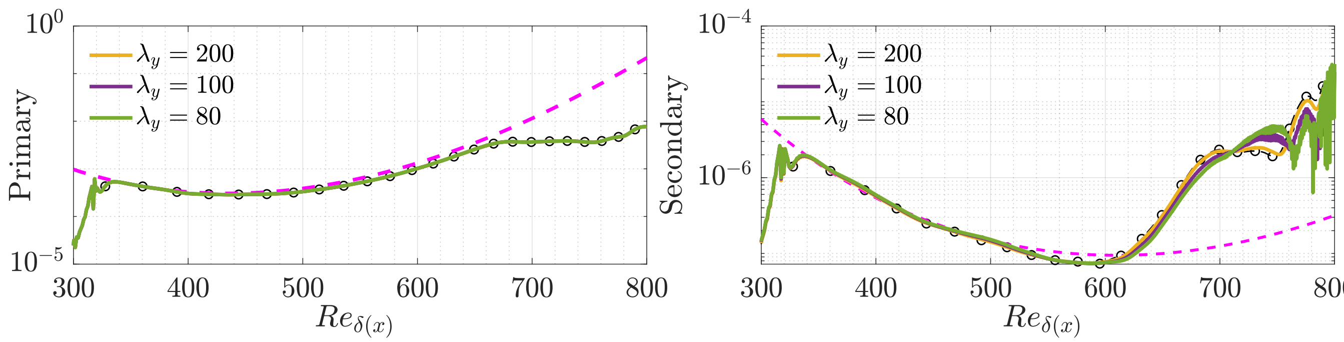
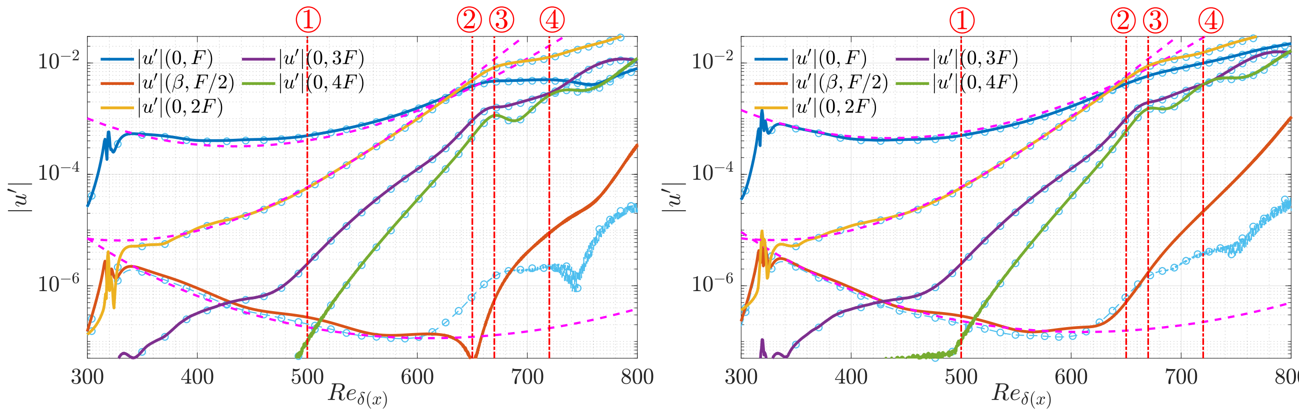


FIGURE 4.21: Primary (left) and subharmonic (right) instability growth in 3D simulations with different spanwise wave length. (- - -): LST prediction. (-∘-): 2D DNS results with $F_1 = 100 \times 10^{-6}$, $A_1 = 10^{-3}$, $F_2 = 50 \times 10^{-6}$ and $A_2 = 5 \times 10^{-6}$.

Afterwards, one simulation with fine mesh is conducted with $\lambda_y = 50$. The exact values of parameters can be found in table A.1 with the name **3D VdW REF**. The first results are computed with fixed y criterion (d) in table 2.3 and are shown in figure 4.22a. Comparing to the 2D simulation results using the same criterion, the secondary growth seems to be delayed. However, when we analyze further this case with more data, applying the maximum y criterion (g), the secondary instability growth is not so significantly delayed (shown in figure 4.22b). For discussing deeper into this observation, instability curves using criterion (d)



(A) Amplitudes using criterion (d)

(B) Amplitudes using criterion (g)

FIGURE 4.22: Amplitude of different harmonics using criterion (d) and (g) in table 2.3. (- - -): LST prediction. (- ⊖ -): 2D DNS results with $A_1 = 10^{-3}$. (-·-·): position markers. ① Re=500, ② Re=650, ③ Re=670, ④ Re=720

with several different normal positions are compared with the ones using criterion (g) in figure 4.23. For the

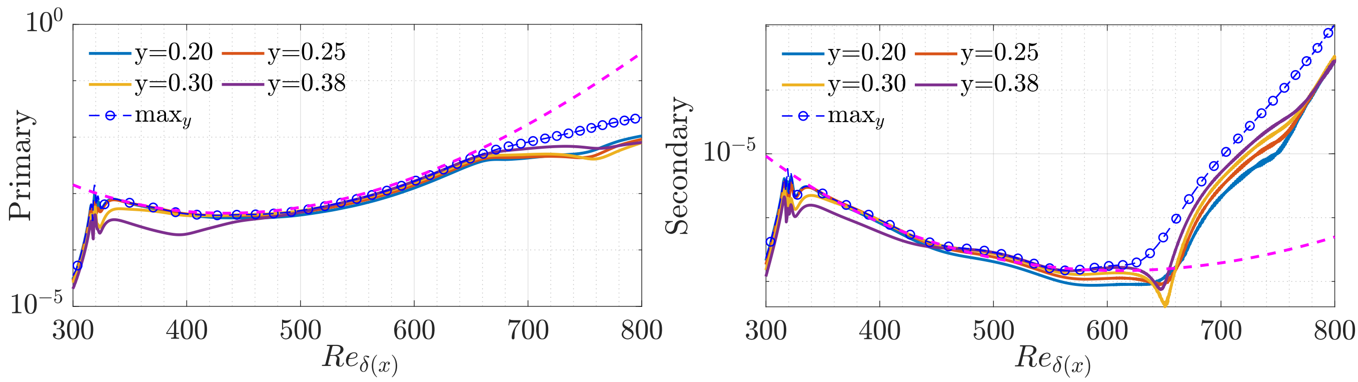
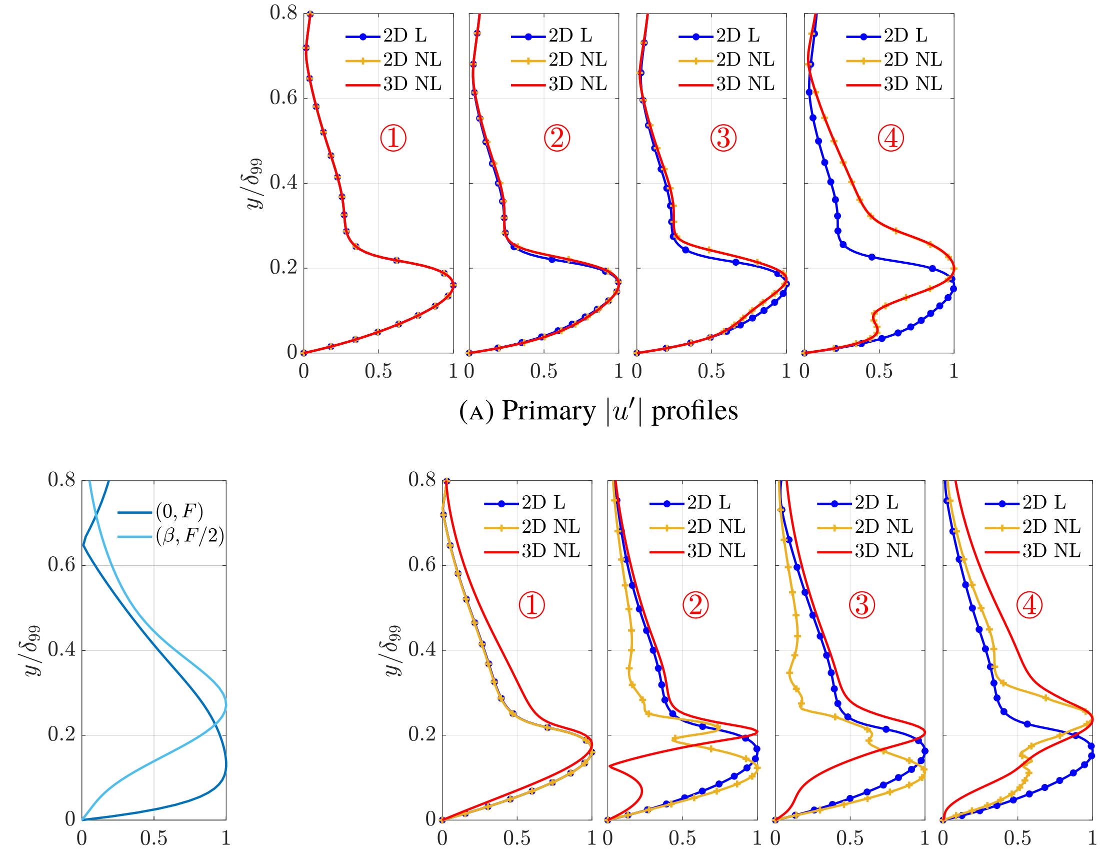


FIGURE 4.23: Primary (left) and subharmonic (right) instability growth using criterion (d) with different y position, compared with criterion (g). (- - -): LST prediction.

linear part of both primary and secondary instabilities, the evolution of amplitude depends little on criteria, but when non-linearity enters into the system, difference between criteria (d) and (g) is much greater and tuning the value of fixed wall normal position can no longer improve the consistency between these two criteria. More precisely, the criterion (g) gives larger value after the appearance of secondary instability growth, and this can be due to the distortion in fluctuation profiles. Keeping this in mind, we would like to look into the fluctuation profiles of $|u'|$ for the next step.

The same method in 2D case is applied to the streamwise velocity fluctuation $|u'|$, and the primary and secondary profiles are plotted at four streamwise position marked in figure 4.22. Figure 4.24 shows the profiles found in this 3D VdW simulation with 2D VdW simulation and ideal gas H-type profiles. For the primary $|u'|$ profiles (figure 4.24a), 2D and 3D VdW results perfectly superimposed, while greater difference can be observed in secondary instability curves. The position ① again presents where neither secondary growth nor non-linear effects enters, and there is little difference between profiles of the three simulations. The secondary growth curve of criterion (d) has a drop at position ②, and we can clearly see a minimum in the profile at this position which explains the growth curve behavior. At position ③, the growth of primary mode starts to deviate from linear prediction but still not much, and the profile also support this statement. Also at this position, we can observe that the maximum of the secondary instability profile shifts upward thus explains the larger value given by criterion (g) than (d). It is also at this position that the secondary profile of $|u'|$ has some similarity to the 3D ideal gas profile. It is possible that 2D



(A) Primary $|u'|$ profiles

Ideal gas $|u'|$ profiles

(B) Subharmonic $|u'|$ profiles

FIGURE 4.24: $|u'|$ profiles of 2D and 3D VdW simulations with $A_1 = 10^{-3}$ at the four positions marked in 4.22b. Values are normalized by the maximum value of each profile. 2D L: 2D DNS results with $A_1 = 5 \times 10^{-8}$. 2D NL: 2D DNS results with $A_1 = 10^{-3}$. The ideal gas curves are derived by the simulation in section 3.2.

fluctuation profiles are not accurate because the 2D Mode II at subharmonic frequency is mixed with the subharmonic secondary mode, but this should be discussed more rigorous with secondary stability analysis. At the last position, non-linearity plays a more important part but the secondary profile still keeps similar shape as the one at position ③, it is hard to conclude whether the growth of subharmonic instability is related to the secondary growth or the non-linear effects brought by higher harmonics at this position and afterwards, but the growth is evidently larger than the 2D case.

After completing the simulation with $\lambda_y = 50$ and fine mesh, we tried to decrease the wave length to 20, but the program crashes even with the fine mesh. As we cannot afford time and computational power to refine more the mesh, there is no result for smaller spanwise wave numbers. However, it still shows that 3D effects with small spanwise wave length can destabilize the flow.

Chapter 5

Conclusion

This study was initiated with the goal of finding the secondary instability growth of the newly discovered unstable mode named Mode II in a flat-plate boundary layer of supercritical fluids. Direct numerical simulation (DNS) serves as a powerful tool to analyze the non-linear effects which are hard to predict with analytical approaches. After using an in-house DNS program in `FORTRAN` to generate data, we then carried out post-processing in `MATLAB` to identify instability behaviors in the flow.

Firstly, 2D and 3D simulations are conducted for incompressible isothermal boundary layer flows with small perturbation. In this part, different methods of instability identification are implemented and compared to determine the ones to be used throughout the whole study. The results show a good consistency with the prediction of linear stability theory (LST), thus validate the postprocessing method. Next, Van der Waals (VdW) equation of state was introduced to model the real gas effects of supercritical fluids. The first 2D simulations of a transcritical boundary layer recovered the LST prediction of Mode II, validated the numerical settings used in the transcritical case, and allowed us to continue the study with higher forcing amplitudes which can trigger secondary instability growth. Finally, 3D real gas simulations were performed and the results helped us quantitatively understand the changes brought by expending from 2D to 3D and the influence of the spanwise wave number of secondary mode.

The outcomes have confirmed the existence of secondary instability growth of Mode II. Apart from the subharmonic mode, significant effects of non-linearity and interactions between higher harmonics are also observed in the transcritical boundary layer, which is not the case for incompressible and isothermal boundary layer flows. The study of spectra, mean and fluctuation profiles confirmed further the existence of non-linear effects, and as a consequence, the amplitude variations captured by different criteria are more different, therefore the criteria should be chosen more carefully. Compared to the subharmonic mode, the amplitudes of higher harmonics increase much faster. Even though a stronger growth of the subharmonic mode was observed in 3D real gas simulations with smaller wave length, it still increases slower than the higher harmonics. For this reason, the transition to turbulence may be less likely to follow the path via secondary instability.

Apart from the conclusions and comments listed above, the simulation results raised other questions about the instability behaviors beyond Mode II and opened several possible perspectives for future investigations. The study of subharmonic secondary instability concentrated mainly on frequency $F = 100 \times 10^{-6}$. As the primary growth changes with frequency, there may be a better choice of primary frequency with larger secondary growth and smaller higher harmonics growth. Apart from that, the study about interactions between modes with different frequencies also shows the worth of tuning forcing parameters and investigating into the resonance interactions. Besides, the results show that, when the forcing amplitude is large, the blowing/suction forcing used in DNS can trigger other harmonics with small but not negligible amplitudes, which are amplified by the large growth rate of Mode II in the downstream. When a better forcing is implemented such that other harmonics do not appear, the study of the secondary instability can be more rigorous and this is one of the possible directions for future work.

Appendix A

DNS simulations dataset

Case	Ma	Ec	Pr	S_f	L_y	L_z	Re_{Start}	Re_{End}	n_x	n_y	n_z	$L_{sp,in}$	$L_{sp,out}$	$L_{sp,fs}$	$Re_{Perturb}$	$L_{Perturb}$	$F_1 \times 10^{-6}$	$F_2 \times 10^{-6}$	A_1	A_2
2D IG REF	0.36	0.05	0.75	4	20	1	150	850	1000	200	1	20	20	1	250	20	100	-	5×10^{-7}	0
2D IG F33	0.36	0.05	0.75	4	20	1	600	1800	2500	200	1	20	20	1	680	10	33	-	5×10^{-7}	0
3D IG H-type	0.20	0.02	0.75	4	20	9.63	$\sqrt{10^5}$	700	800	200	30	20	20	1	415.8	10	124	62	2.25×10^{-3}	7.5×10^{-5}
2D VdW REF	0.42	0.05	1	6	40	1	250	900	2000	300	1	28	150	5	320	5	100	-	5×10^{-10}	0
2D VdW RDU	0.42	0.05	1	6	40	1	250	900	1000	200	1	28	150	5	320	5	100	50	10^{-3}	5×10^{-6}
3D VdW RDU	0.42	0.05	1	6	40	80	250	900	1000	200	30	28	150	5	320	5	100	50	10^{-3}	5×10^{-6}
3D VdW REF	0.42	0.05	1	6	40	50	250	900	2000	300	30	28	150	5	320	5	100	50	10^{-3}	5×10^{-6}

TABLE A.1: Physical and numerical parameter values of referential DNS cases. Ma: the Mach number. Ec: the Eckert number. Pr: the Prandtl number. S_f: the grid streching factor. L_y: the height of the domain. L_z: the spanwise length of the domain. Re_{Start} and Re_{End}: the Reynolds number defining start and end streamwise position of the domain. n_x: number of grid points in streamwise direction. n_y: number of grid points in wall normal direction. n_z: number of grid points in spanwise direction. $L_{sp,in}$, $L_{sp,out}$ and $L_{sp,fs}$: sponge length at inlet, outlet and on the top of the domain. $Re_{Perturb}$: Reynolds number defining the center of the perturbation. $L_{Perturb}$: perturbation length. F_1 and F_2: dimensionless frequencies of the forcing, related to ω_1 and ω_2 in equation (2.29). A_1 and A_2: forcing amplitudes related to forcing F_1 and F_2.

Appendix B

Computational domain validation for 2D ideal gas case

B.1 Mesh convergence

In this part we show the validation of the mesh used in 2D ideal gas reference case, results with different mesh are compared. All results are calculated by criterion (c) in table 2.3.

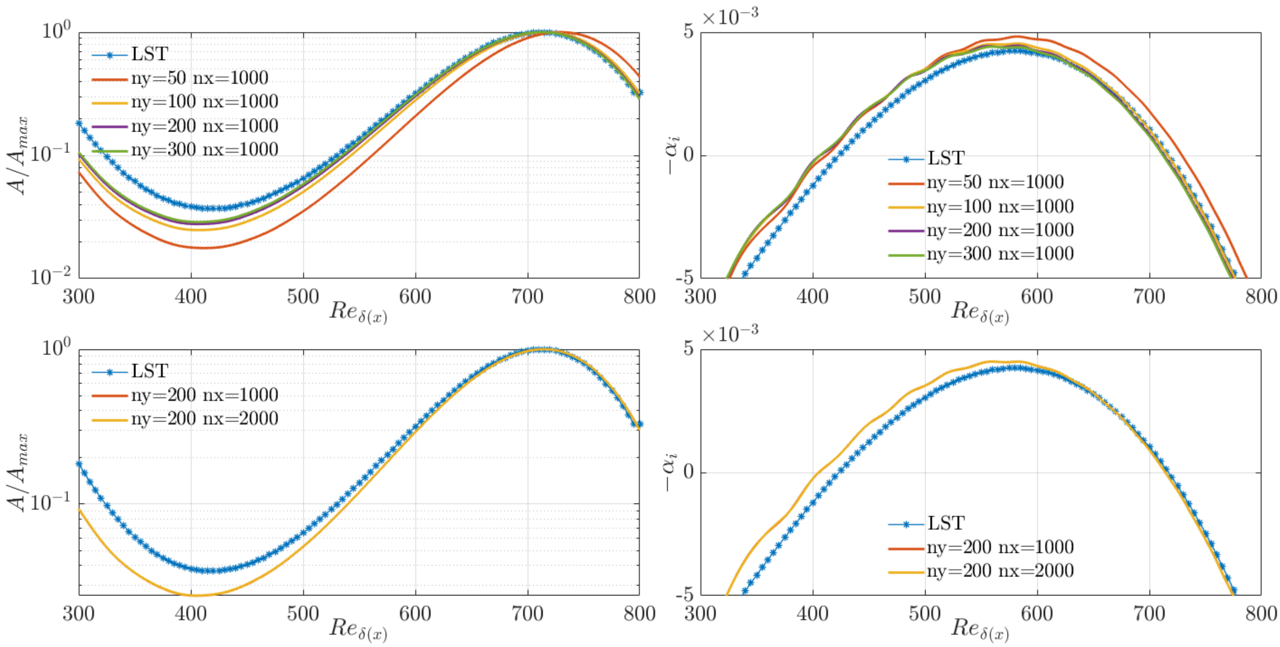


FIGURE B.1: Mesh dependency with criterion (c) for 2D reference case. Left: amplitude, right: related growth rate. Top: mesh dependency in wall normal direction, bottom: mesh dependency in streamwise direction

In the first two charts on the top of figure B.1, the number of grid points in the streamwise direction is fixed to 1000 and the number of grid points in the wall normal direction varies from 50 to 300. The comparison shows that using 200 points in the wall normal direction is enough to get the converged growth rate. In the other two charts, the number of grid points in the wall normal direction is fixed to 200 and we compare the results with respectively 1000 and 2000 grid points in the streamwise direction. We can conclude that the $(200 * 1000)$ mesh is enough to find the converged result.

B.2 Validation of the domain height

Although sponge and correct boundary conditions are implemented in DNS simulations, the height of the domain still should be large enough so that the upper boundary do not influence flow's behaviors in the boundary layer.

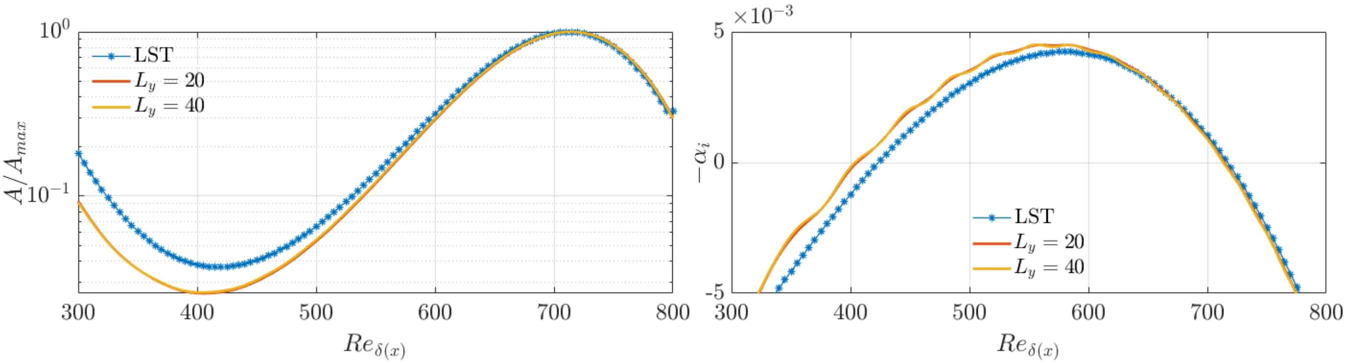
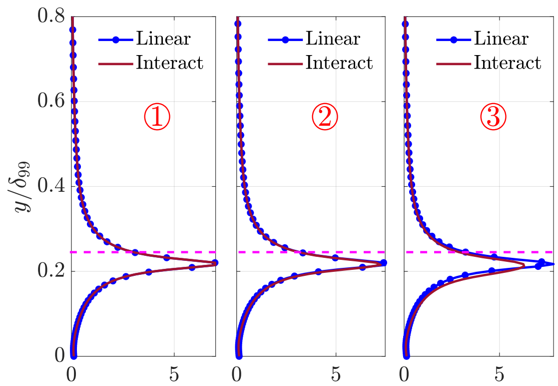


FIGURE B.2: Validation of the domain height in 2D case with criterion (c). Left: amplitude, right: related growth rate.

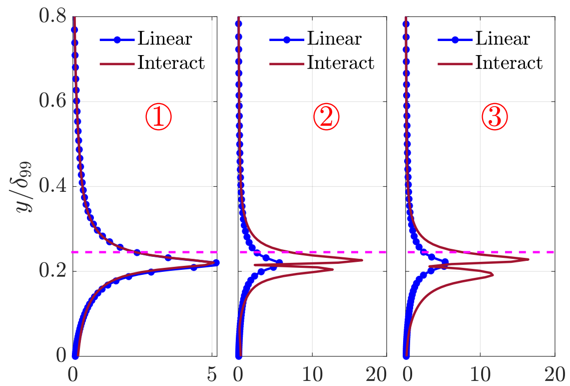
Ideally, same grids in the wall normal direction should be used with different domain height in order to determine whether L_y is correctly chosen. However, because of the stretching function used to define grid, it's difficult to match grids with different domain height. In order to verify the choice of domain height $L_y = 20$ is enough, we compare the converged result with the converged result of a higher domain ($L_y = 40$) although the grids are not matching. Results are shown in figureB.2. Nearly no difference is observed between those results and the choice $L_y = 20$ is validated.

Appendix C

Primary and secondary profiles of $|\rho'|$ and $|v'|$ for 2D VdW simulations

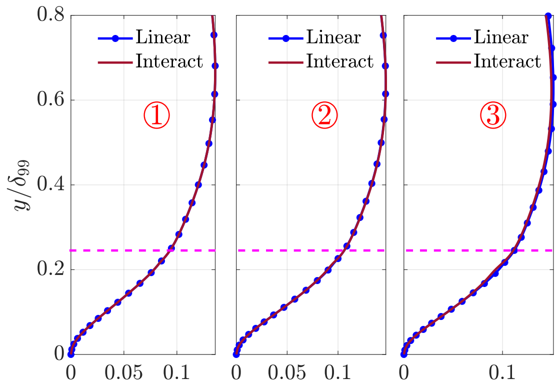


(A) Primary $|\rho'|$ profiles

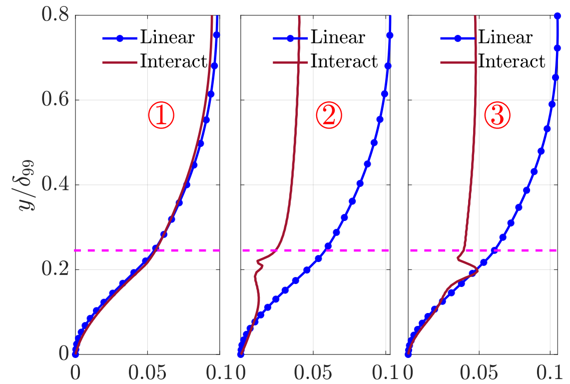


(B) Subharmonic $|\rho'|$ profiles

FIGURE C.1: $|\rho'|$ profiles of VdW simulation with $A_1 = 10^{-4}$ at the three positions marked in 4.9a. Values are normalized by the maximum value of $|u'|$ profile at each position. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.

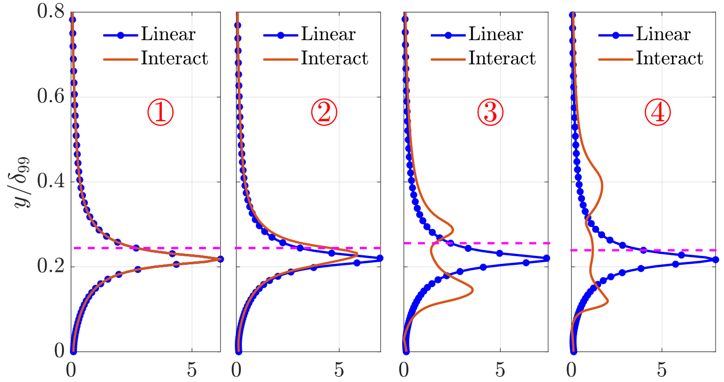


(A) Primary $|v'|$ profiles

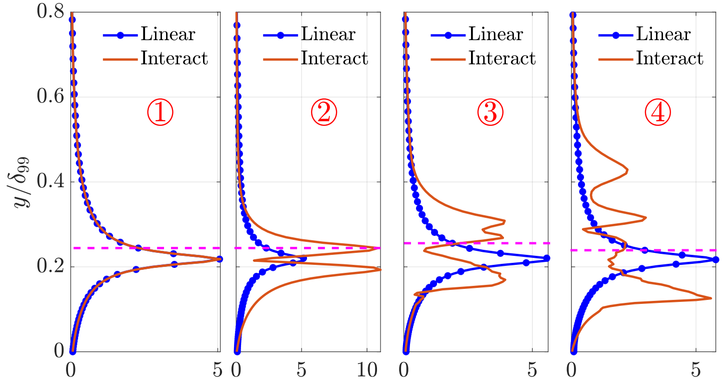


(B) Subharmonic $|v'|$ profiles

FIGURE C.2: $|v'|$ profiles of VdW simulation with $A_1 = 10^{-4}$ at the three positions marked in 4.9a. Values are normalized by the maximum value of $|u'|$ profile at each position. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.

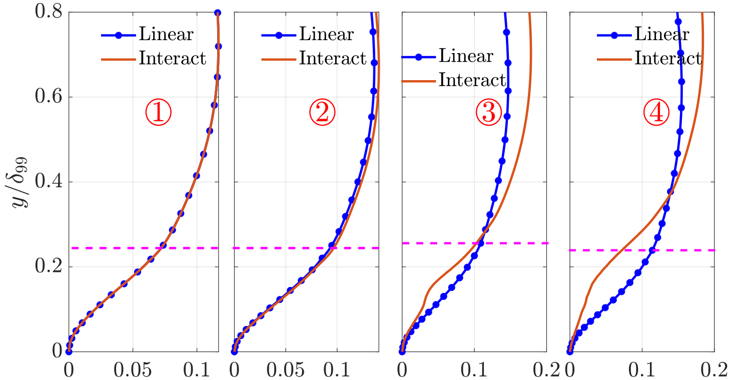


(A) Primary $|\rho'|$ profiles

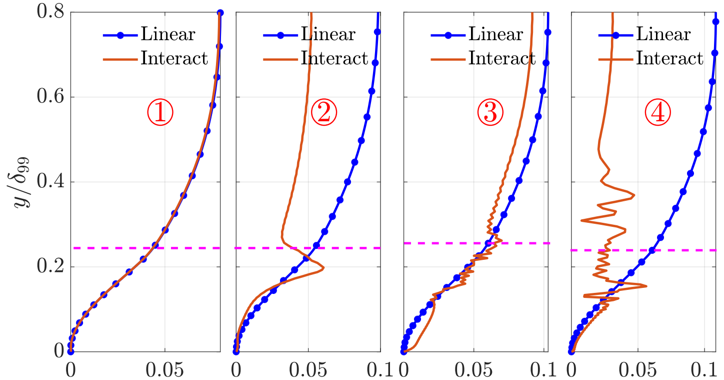


(B) Subharmonic $|\rho'|$ profiles

FIGURE C.3: $|\rho'|$ profiles of VdW simulation with $A_1 = 10^{-4}$ at the three positions marked in 4.9a. Values are normalized by the maximum value of $|u'|$ profile at each position. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.



(A) Primary $|v'|$ profiles



(B) Subharmonic $|v'|$ profiles

FIGURE C.4: $|v'|$ profiles of VdW simulation with $A_1 = 10^{-4}$ at the three positions marked in 4.9a. Values are normalized by the maximum value of $|u'|$ profile at each position. (—•— Linear): DNS results with $A_1 = 5 \times 10^{-8}$. (- - -): position where $T = T_{pc}$.

Bibliography

B. Bugeat, P. C. Boldini, and R. Pecnik. On the new instability mode in the boundary layer flow of supercritical fluids. In *12th International Symposium on Turbulence and Shear Flow Phenomena*, Osaka, Japan, July 2022.

F. Charru. *Hydrodynamic instabilities*. Cambridge texts in applied mathematics. Cambridge University Press, Cambridge ;, 2011. URL https://doi.org/10.1017/CBO9780511975172.

X. Cheng and T. Schulenberg. *Heat transfer at supercritical pressures: literature review and application to an HPLWR*, volume 6609. FZKA Karlsruhe, Germany, 2001.

A. Craik. Second order resonance and subcritical instability. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, 343(1634):351–362, 1975.

H. Fasel and U. Konzelmann. Non-parallel stability of a flat-plate boundary layer using the complete Navier-Stokes equations. *Journal of Fluid Mechanics*, 221:311–347, Dec. 1990. ISSN 1469-7645, 0022-1120. doi: 10.1017/S0022112090003585. Publisher: Cambridge University Press.

M. Gaster. On the effects of boundary-layer growth on flow stability. *Journal of Fluid Mechanics*, 66(3): 465–480, Nov. 1974. ISSN 0022-1120, 1469-7645. doi: 10.1017/S0022112074000310.

S. Grossmann. Instability without instability? In *Nonlinear Physics of Complex Systems*, pages 10–22. Springer, 1996.

J. He, R. Tian, P. Jiang, and S. He. Turbulence in a heated pipe at supercritical pressure. *Journal of Fluid Mechanics*, 920, 2021.

T. Herbert. Secondary instability of plane channel flow to subharmonic three-dimensional disturbances. *The Physics of Fluids*, 26(4):871–874, 1983.

T. Herbert. Secondary instability of plane shear flows—theory and application. In *Laminar-Turbulent Transition*, pages 9–20. Springer, 1985.

T. Herbert. Secondary instability of boundary layers. *Annual review of fluid mechanics*, 20(1):487–526, 1988. Publisher: Annual Reviews 4139 El Camino Way, PO Box 10139, Palo Alto, CA 94303-0139, USA.

T. Herbert, F. P. Bertolotti, and G. R. Santos. Floquet analysis of secondary instability in shear flows. In *Stability of time dependent and spatially varying flows*, pages 43–57. Springer, 1987.

J. A. Jossi, L. I. Stiel, and G. Thodos. The viscosity of pure substances in the dense gaseous and liquid phases. *AIChE Journal*, 8(1):59–63, 1962.

Y. S. Kachanov and V. Y. Levchenko. The resonant interaction of disturbances at laminar-turbulent transition in a boundary layer. *Journal of Fluid Mechanics*, 138:209–247, 1984. doi: 10.1017/S0022112084000100.

S. Kawai. Direct numerical simulation of transcritical turbulent boundary layers at supercritical pressures with strong real fluid effects. In *54th AIAA Aerospace Sciences Meeting*, page 1934, 2016.

P. S. Klebanoff, K. D. Tidstrom, and L. M. Sargent. The three-dimensional nature of boundary-layer instability. *Journal of Fluid Mechanics*, 12(1):1–34, Jan. 1962. ISSN 1469-7645, 0022-1120. doi: 10.1017/S0022112062000014. Publisher: Cambridge University Press.

S. E. Krist and T. A. Zang. Numerical simulation of channel flow transition, resolution requirements and structure of the hairpin vortex, Apr. 1987.

L. Landau and E. M. Lifshits. Mechanics of continuous media. *Gostekhizdat, Moscow*, 81, 1954.

L. Lees and C.-C. Lin. *Investigation of the stability of the laminar boundary layer in a compressible fluid*. Number 1115 in Technical note. National Advisory Committee for Aeronautics, 1946.

Y. Liu, Y. Wang, and D. Huang. Supercritical co2 brayton cycle: A state-of-the-art review. *Energy*, 189: 115900, 2019.

L. M. Mack. Boundary-layer linear stability theory. AGARD report 709, California Inst of Tech Pasadena Jet Propulsion Lab, 1984.

M. V. Morkovin. Critical evaluation of transition from laminar to turbulent shear layers with emphasis on hypersonically traveling bodies. Technical report, MARTIN MARIETTA CORP BALTIMORE MD RESEARCH INST FOR ADVANCED STUDIES, Mar. 1969. Section: Technical Reports.

R. Pecnik and A. Patel. Scaling and modelling of turbulence in variable property channel flows. *Journal of Fluid Mechanics*, 823, 2017.

R. Pecnik, E. Rinaldi, and P. Colonna. Computational Fluid Dynamics of a Radial Compressor Operating With Supercritical CO2. *Journal of Engineering for Gas Turbines and Power*, 134(12), 10 2012. ISSN 0742-4795. doi: 10.1115/1.4007196. URL https://doi.org/10.1115/1.4007196. 122301.

S. Pirozzoli. Generalized conservative approximations of split convective derivative operators. *Journal of Computational Physics*, 229(19):7180–7190, Sept. 2010. doi: 10.1016/j.jcp.2010.06.006.

M. Pizzarelli, F. Nasuti, M. Onofri, P. Roncioni, R. Votta, and F. Battista. Heat transfer modeling for supercritical methane flowing in rocket engine cooling channels. *Applied Thermal Engineering*, 75: 600–607, 2015.

T. Poinsot and S. Lelef. Boundary conditions for direct simulations of compressible viscous flows. *Journal of Computational Physics*, 101(1):104–129, 1992. ISSN 0021-9991. doi: https://doi.org/10.1016/0021-9991(92)90046-2. URL https://www.sciencedirect.com/science/article/pii/0021999192900462.

L. Prandtl. Bemerkungen über die entstehung der turbulenz. *ZAMM-Journal of Applied Mathematics and Mechanics/Zeitschrift für Angewandte Mathematik und Mechanik*, 1(6):431–436, 1921.

M. Raju, D. T. Banuti, P. C. Ma, and M. Ihme. Widom Lines in Binary Mixtures of Supercritical Fluids. *Scientific Reports*, 7(1):3027, June 2017. ISSN 2045-2322. doi: 10.1038/s41598-017-03334-3. Number: 1 Publisher: Nature Publishing Group.

J. Ren, S. Fu, and R. Pecnik. Linear instability of Poiseuille flows with highly non-ideal fluids. *Journal of Fluid Mechanics*, 859:89–125, Jan. 2019a. ISSN 0022-1120, 1469-7645. doi: 10.1017/jfm.2018.815. Publisher: Cambridge University Press.

J. Ren, O. Marxen, and R. Pecnik. Boundary-layer stability of supercritical fluids in the vicinity of the Widom line. *Journal of Fluid Mechanics*, 871:831–864, July 2019b. ISSN 0022-1120, 1469-7645. doi: 10.1017/jfm.2019.348. Publisher: Cambridge University Press.

U. Rist and H. Fasel. Direct numerical simulation of controlled transition in a flat-plate boundary layer. *Journal of Fluid Mechanics*, 298:211–248, 1995. doi: 10.1017/S0022112095003284.

W. Saric and A. Thomas. Experiments on the subharmonic route to turbulence in boundary layers. *Turbulence and chaotic phenomena in fluids*, pages 117–122, 1984.

T. Sayadi, C. W. Hamman, and P. Moin. Direct numerical simulation of complete H-type and K-type transitions with implications for the dynamics of turbulent boundary layers. *Journal of Fluid Mechanics*, 724: 480–509, June 2013. ISSN 0022-1120, 1469-7645. doi: 10.1017/jfm.2013.142. Publisher: Cambridge University Press.

H. Schlichting. Zur enstehung der turbulenz bei der plattenströmung. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1933:181–208, 1933. URL http://eudml.org/doc/59420.

H. Schlichting and K. Gersten. Onset of Turbulence (Stability Theory). In *Boundary-Layer Theory*, pages 415–496. Springer, Berlin, Heidelberg, 2017. ISBN 978-3-662-52919-5. doi: 10.1007/978-3-662-52919-5_15.

P. J. Schmid and D. S. Henningson. *Nonlinear Stability*, pages 153–193. Springer New York, New York, NY, 2001a. ISBN 978-1-4613-0185-1. doi: 10.1007/978-1-4613-0185-1_5. URL https://doi.org/10.1007/978-1-4613-0185-1_5.

P. J. Schmid and D. S. Henningson. Transition to Turbulence. In *Stability and Transition in Shear Flows*, Applied Mathematical Sciences, pages 401–475. Springer, New York, NY, 2001b. ISBN 978-1-4613-0185-1. doi: 10.1007/978-1-4613-0185-1_9.

G. B. Schubauer and H. K. Skramstad. Laminar boundary-layer oscillations and transition on. *Journal of research of the National Bureau of Standards*, 38:251, 1947.

F. Sciortino, P. H. Poole, U. Essmann, and H. E. Stanley. Line of compressibility maxima in the phase diagram of supercooled water. *Physical Review E*, 55(1):727–737, Jan. 1997. doi: 10.1103/PhysRevE.55.727. Publisher: American Physical Society.

C. Shannon. Communication in the presence of noise. *Proceedings of the IRE*, 37(1):10–21, 1949. doi: 10.1109/JRPROC.1949.232969.

N. Sharan and J. Bellan. Investigation of high-pressure turbulent jets using direct numerical simulation. *Journal of Fluid Mechanics*, 922, 2021.

L. I. Stiel and G. Thodos. The thermal conductivity of nonpolar substances in the dense gaseous and liquid regions. *AIChE Journal*, 10(1):26–30, 1964.

G. Stuckert and H. L. Reed. Linear disturbances in hypersonic, chemically reacting shock layers. *AIAA journal*, 32(7):1384–1393, 1994.

W. Tollmien. Über die entstehung der turbulenz. In *Vorträge aus dem Gebiete der Aerodynamik und verwandter Gebiete*, pages 18–21. Springer, 1930.

L. N. Trefethen, A. E. Trefethen, S. C. Reddy, and T. A. Driscoll. Hydrodynamic stability without eigenvalues. *Science*, 261(5121):578–584, 1993.

A. M. Yaglom and U. Frisch. Stability to Finite Disturbances: Energy Method and Landau's Equation. In A. M. Yaglom and U. Frisch, editors, *Hydrodynamic Instability and Transition to Turbulence*, Fluid Mechanics and Its Applications, pages 291–464. Springer, Dordrecht, 2012. ISBN 978-94-007-4237-6. doi: 10.1007/978-94-007-4237-6_4.

J. Y. Yoo. The turbulent flows of supercritical fluids with heat transfer. *Annual review of fluid mechanics*, 45:495–525, 2013.